



Approximation Algorithms for Big Data

Lecture 4: Probabilistic tree embeddings

Spring 2026

Instructor: Tal Wagner

Administrative Things

- HW1 due tonight

Recap

- We covered: Approximate nearest neighbor search
 - Definitions
 - ANN via JL dimension reduction
 - ANN via locality-sensitive hashing (LSH)
 - Construction of Euclidean LSH
 - Practical ANN algorithms:
 - IVF with k-means
 - Graph-based methods: HNSW

Recap Questions

- How did we amplify ANN?

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- How did we amplify ANN?
- Why didn't we need MoM?

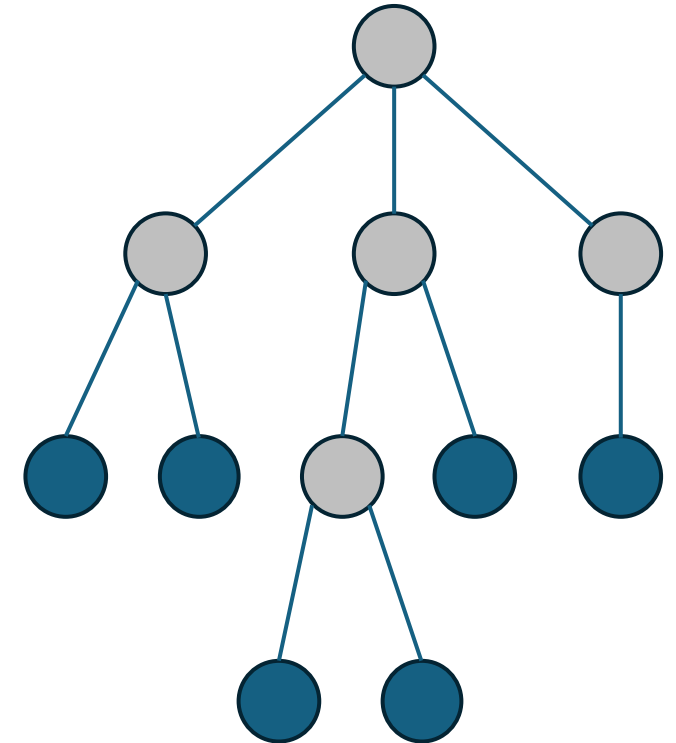
Plan

- **Probabilistic embeddings into trees**
 - Part 1: Background
 - Formal definitions
 - Some background results
 - Part 2: High-dimensional quadtree
 - Construction
 - Distortion analysis
 - Part 3: Applications
 - Approximate nearest neighbor search (ANN)
 - Earth Mover Distance (EMD)

Embedding into Tree Metrics

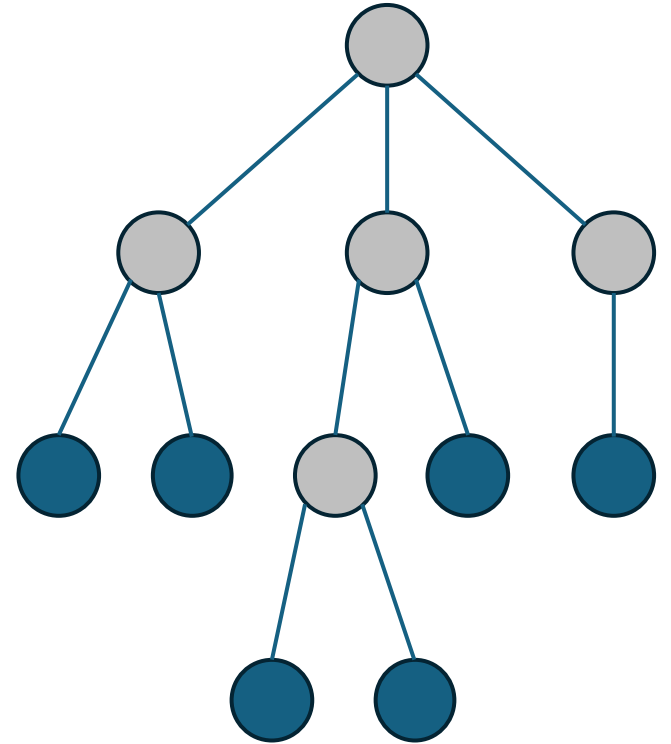
Tree Metrics

- Let T be a weighted tree with n leaves
 - The edges are undirected and have non-negative weights
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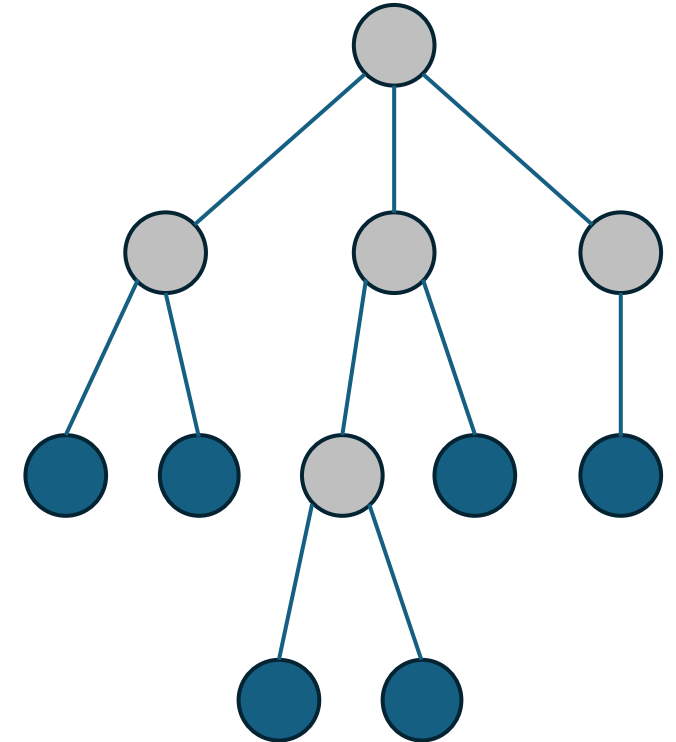
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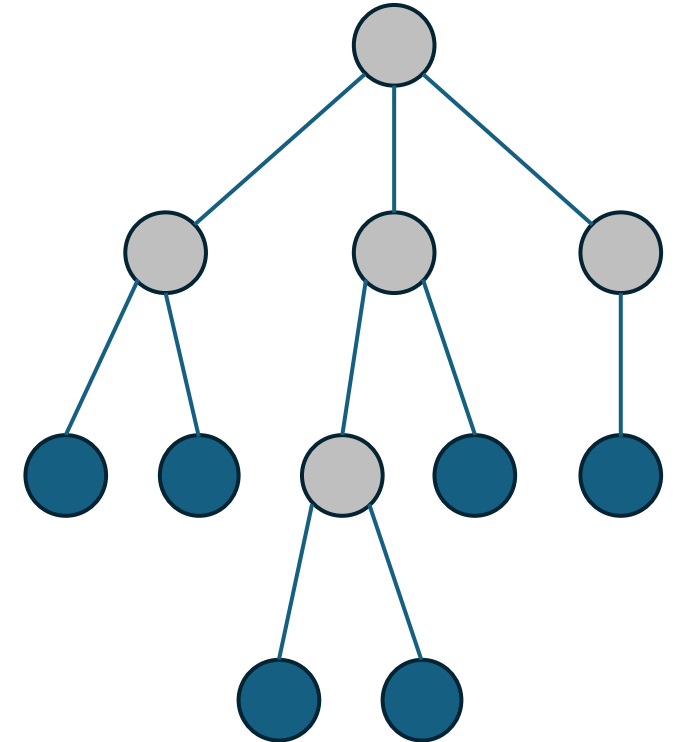
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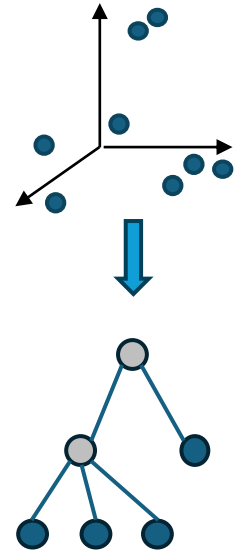
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 - Why is this a metric?



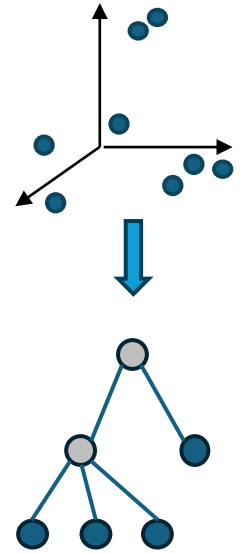
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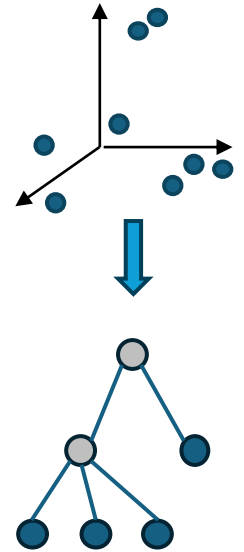
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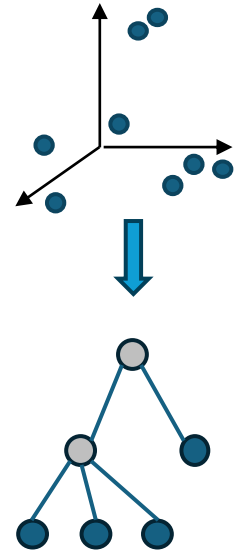
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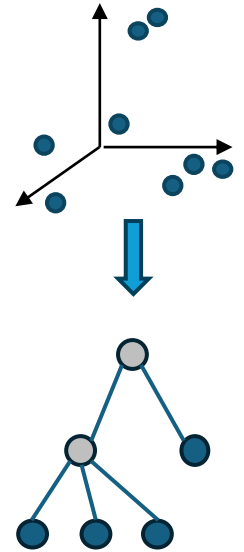
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 - Theorem: Embedding an n -point Euclidean metric into a tree metric might require distortion $\Omega(n)$



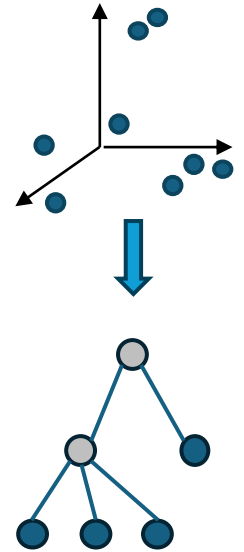
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- **Solution?**



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- **Definition**: Let (X, D) be a metric. Let $b, c > 0$. A probabilistic embedding into tree metrics is a distribution $\mathcal{T}$ over tree metrics such that:
 - We can sample $T \sim \mathcal{T}$ efficiently
 - The leaves of T correspond to the points in X
 - For every $x, y \in T$:
 - $D(x, y) \leq b \cdot T(x, y)$ deterministically, for every supported T
 - $\mathbb{E}_{T \sim \mathcal{T}}[T(x, y)] \leq c \cdot D(x, y)$
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- Developed in [Karp '89; Alon, Karp, Peleg, West '95; Bartal '96]

Probabilistic Tree Embedding: Results

- Bartal (1996):
 - **Theorem**: Every n -point metric space admits a probabilistic tree embedding with distortion $O(\log^2 n)$
 - **Theorem**: Lower bound: $\Omega(\log n)$
- Bartal (1998): Improved upper bound to $O(\log n \log \log n)$
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- All the constructions results yield **hierarchically well-separated trees (HSTs)**

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- Approximation:
 - **Before FRT'03**: $O(\log^2 n)$ and then $O(\log^{1.5} n)$ [Feige & Krauthgamer '00]
 - Building on [Leighton & Rao '99] and then [Arora-Rao-Vazirani'04] Sparsest Cut approximation
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- There are **deep connections** between metric spaces and metric embedding and combinatorial graph problems

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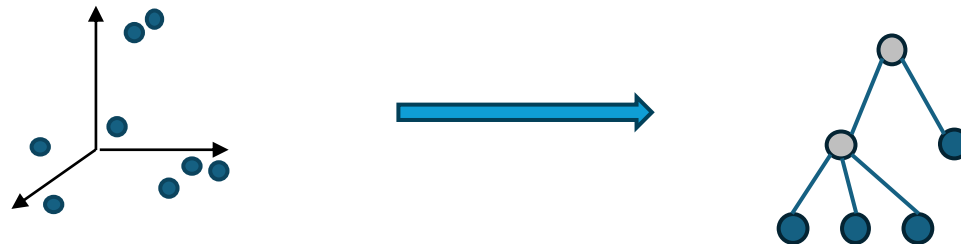
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 - **How come the same algorithm works for both $\ell_2 \rightarrow \ell_2$ and $\ell_2 \rightarrow \ell_1$?**
- **Upshot**: we can assume essentially w.l.o.g. that our Euclidean metric is already embedded as an ℓ_1 metric
- For the probabilistic trees we will see, it will be more convenient to work with ℓ_1

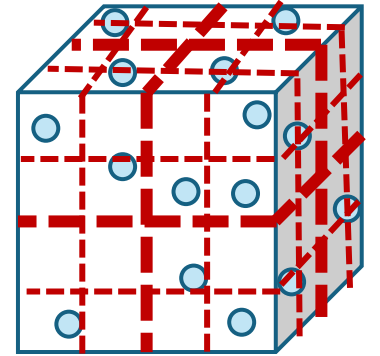
Embedding into High-Dimensional Quadtree

- Setting: we have a set X of n points in $\mathbb{R}^d$, with coordinates bounded in $[-\Delta, \Delta]$
 - It is standard to assume that coordinates are polynomially bounded: $\Delta = n^{O(1)}$
 - By JL we will assume that $d = O(\log n)$
 - We're going for polylog distortion anyway, so we might as well ignore ε and set it to some constant
 - By normalization: the minimum distance in X is 1
 - What is the diameter?

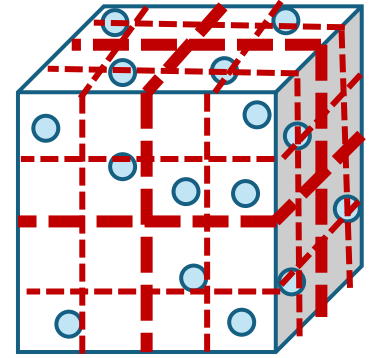


High-Dimensional Quadtree

- **Construction:**
 - We impose on the points a [hierarchy of nested grids](#)
 - The side-length of each grid is half that of the previous grid

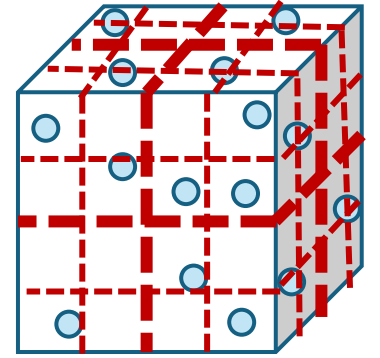


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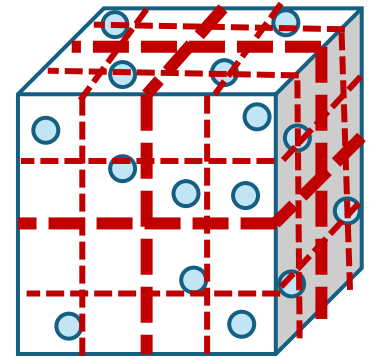
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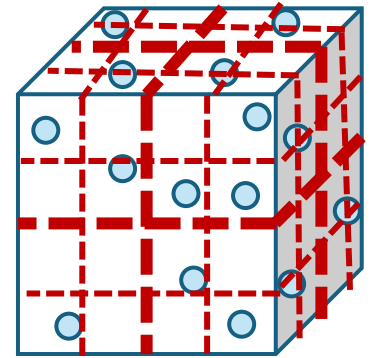
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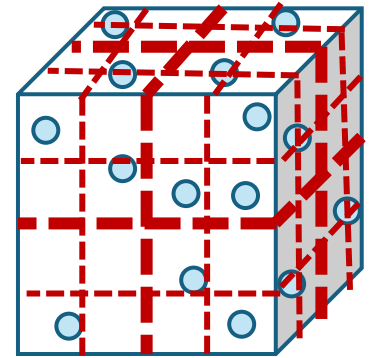
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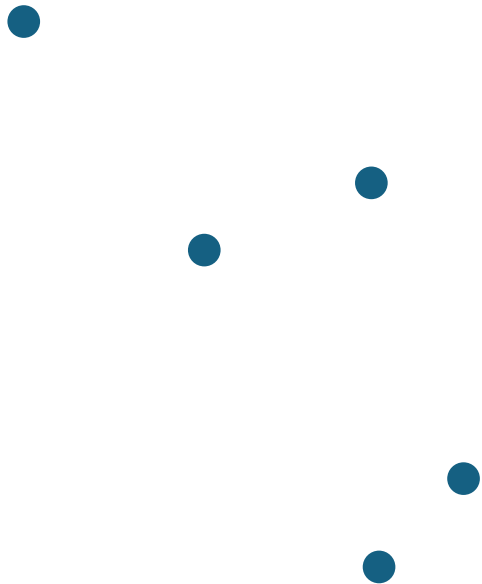
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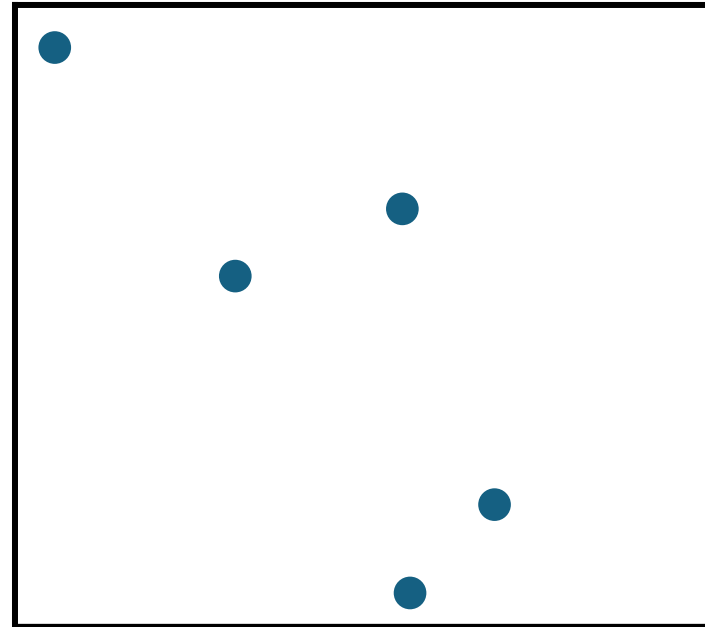


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 - We will build the tree for $L := \log_2(2\Delta d)$ levels (why this number?)

High-Dimensional Quadtree: Construction

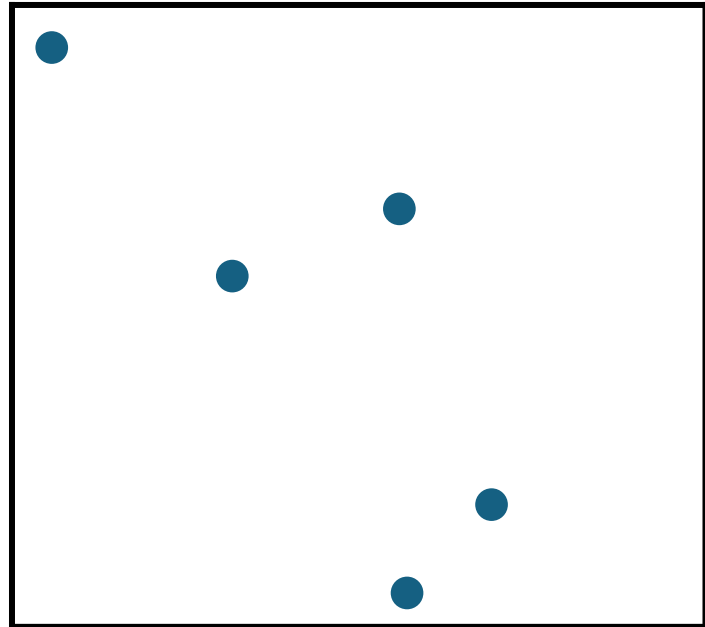


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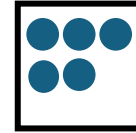


Side
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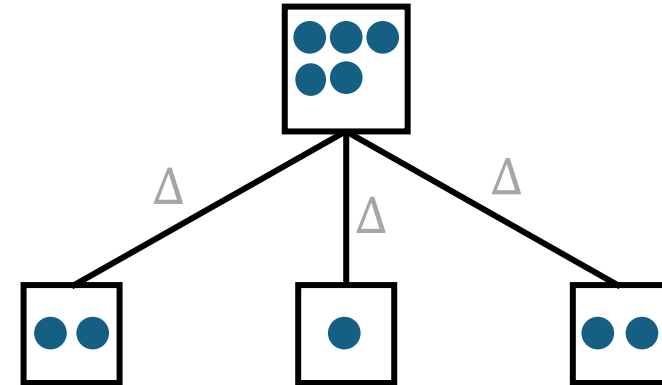
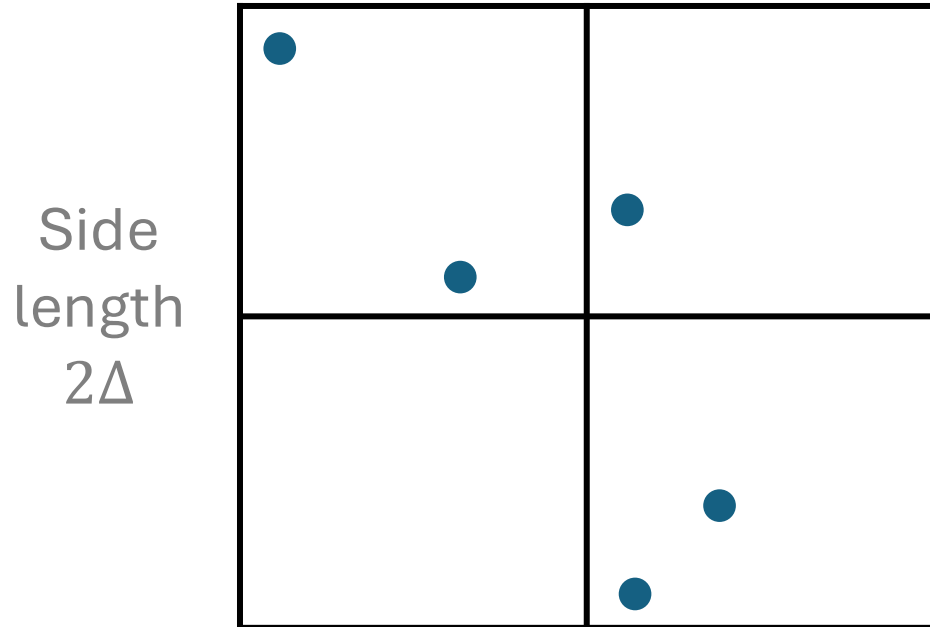
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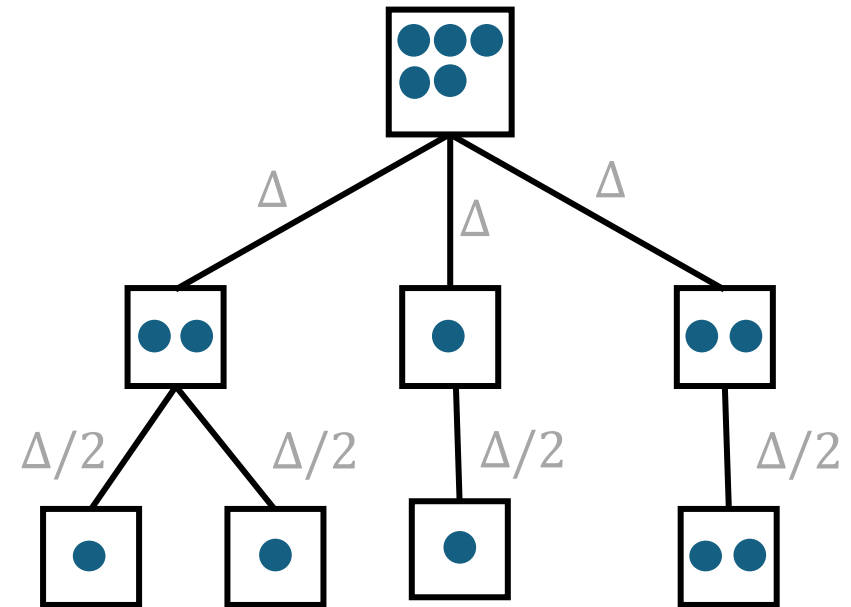
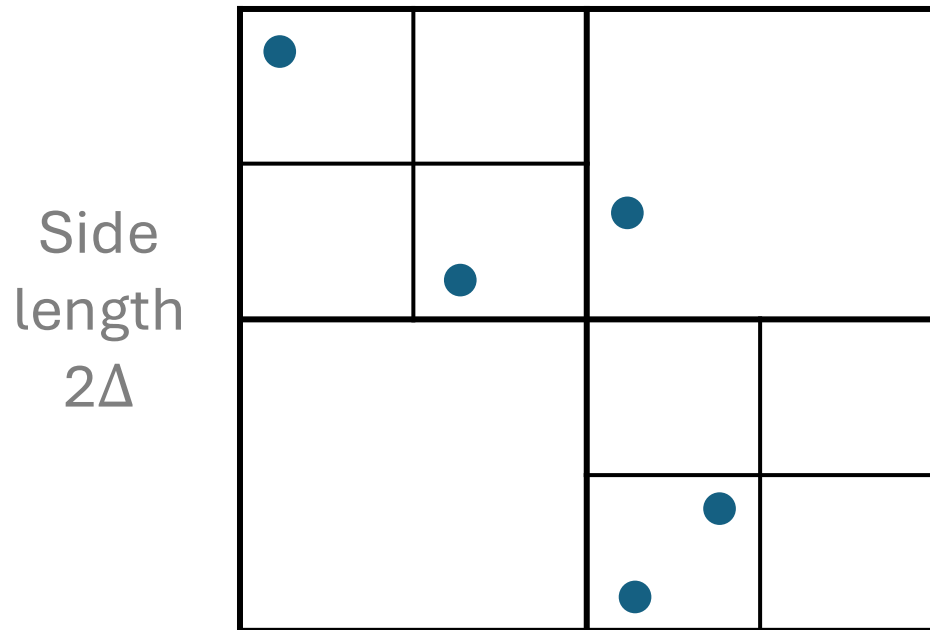
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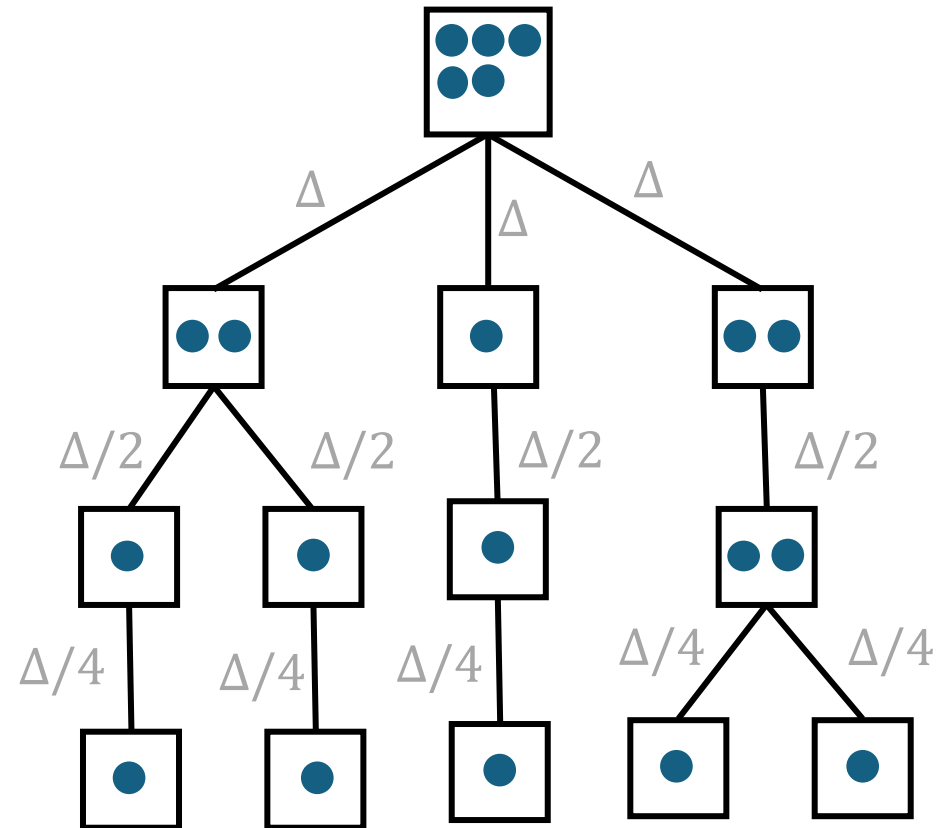
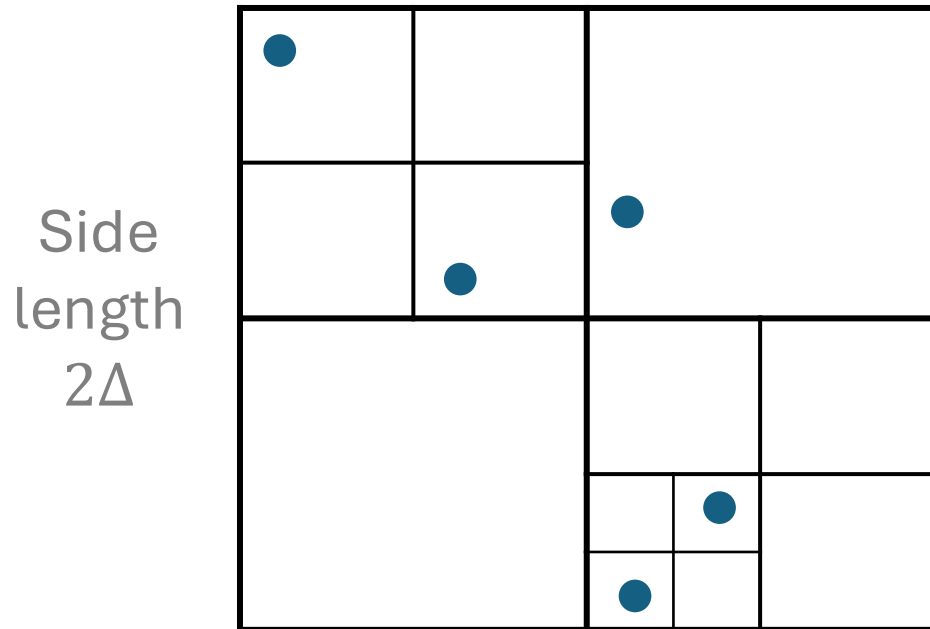
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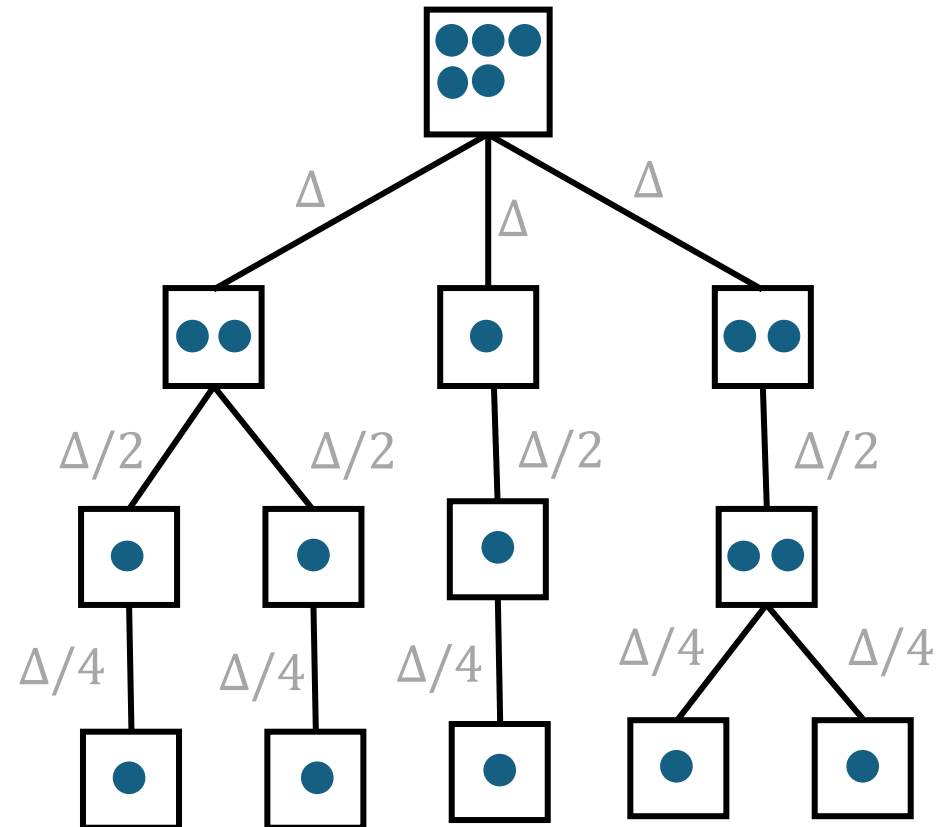
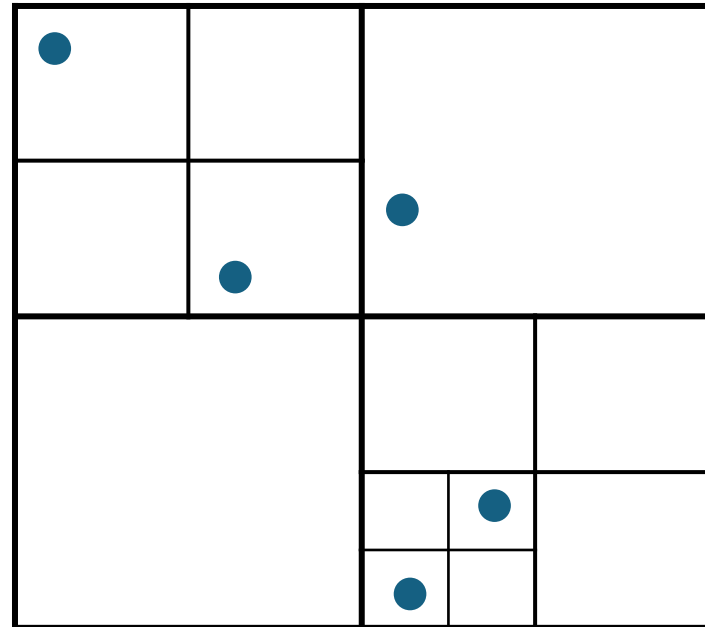


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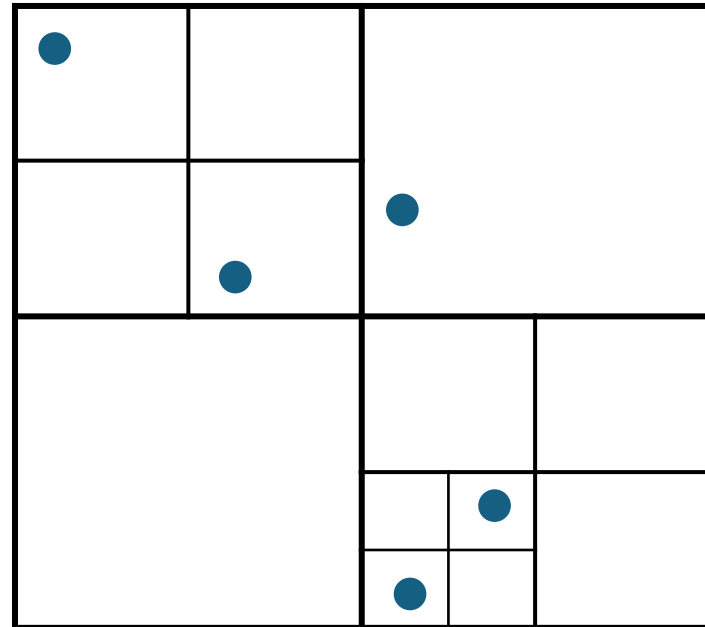
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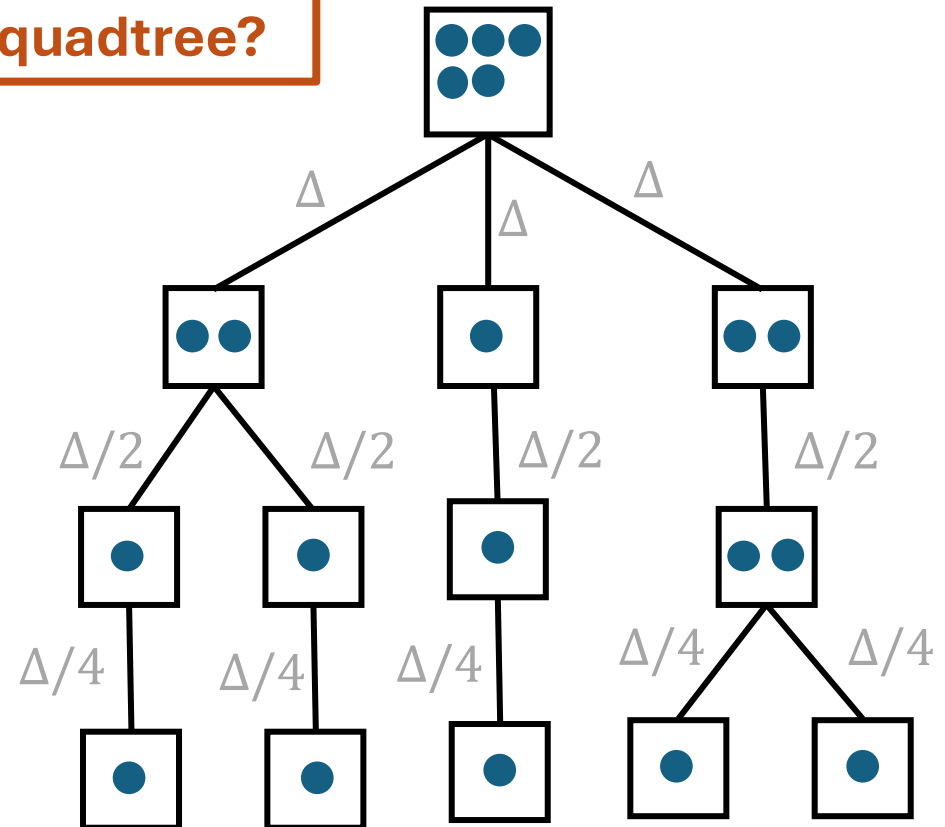
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What is the
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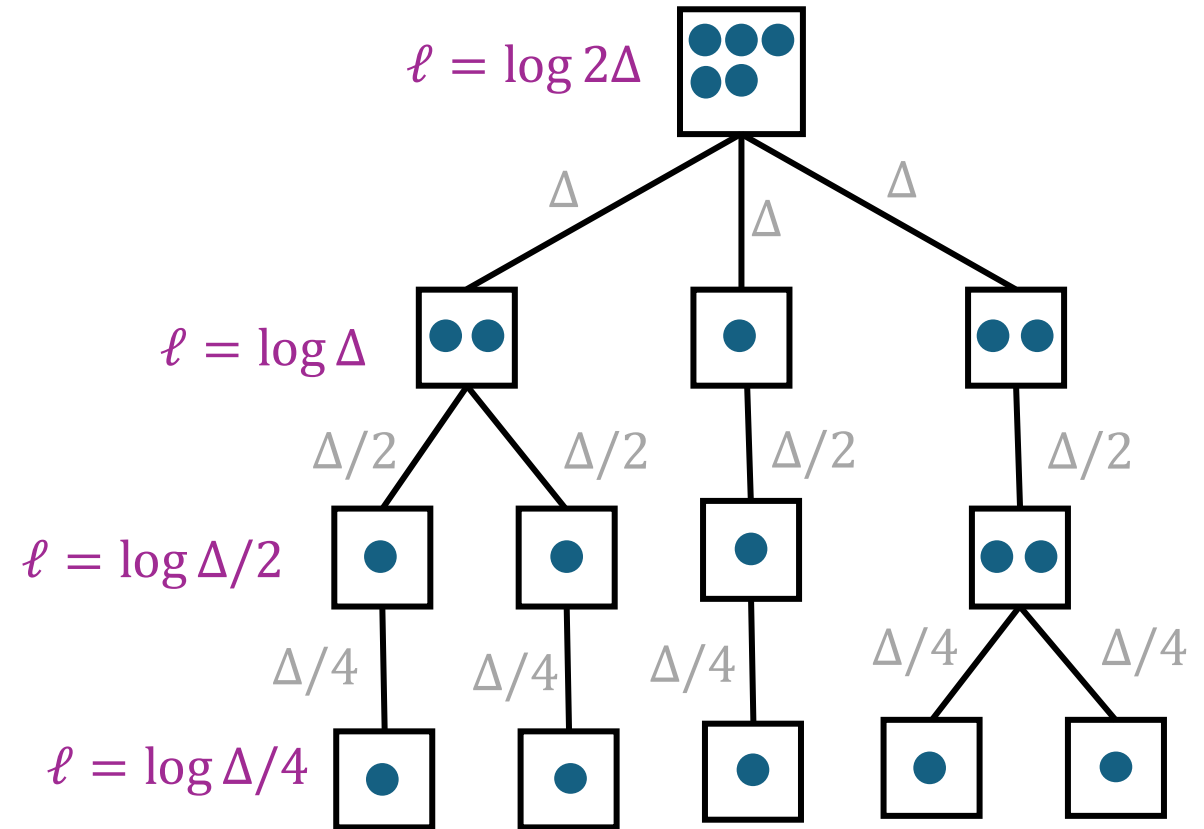


High-Dimensional Quadtree: Distortion

- **Theorem:** The probabilistic embedding into a high-dimensional quadtree with a randomly shifted hierarchical grid satisfies for all $x, y \in X$:
 - $\|x - y\|_1 \leq O(d) \cdot T(x, y)$, deterministically
 - $\mathbb{E}_T[T(x, y)] \leq O(\log(\Delta d)) \cdot \|x - y\|_1$
- Therefore, the distortion is $O(d \log(\Delta d))$
- Recall our natural assumptions: $d = O(\log n)$, $\Delta = n^{O(1)}$
- Under these assumptions the distortion is $\log^{O(1)} n$

Level Numbering

- There are $\log(2\Delta d)$ levels in the tree
- We number them $\ell = \log(1/d), \dots, \log(2\Delta)$
- This way:
 - A node at level ℓ corresponds to hypercubes with side-length 2^ℓ
 - An edge connecting levels ℓ and $\ell + 1$ has weight 2^ℓ

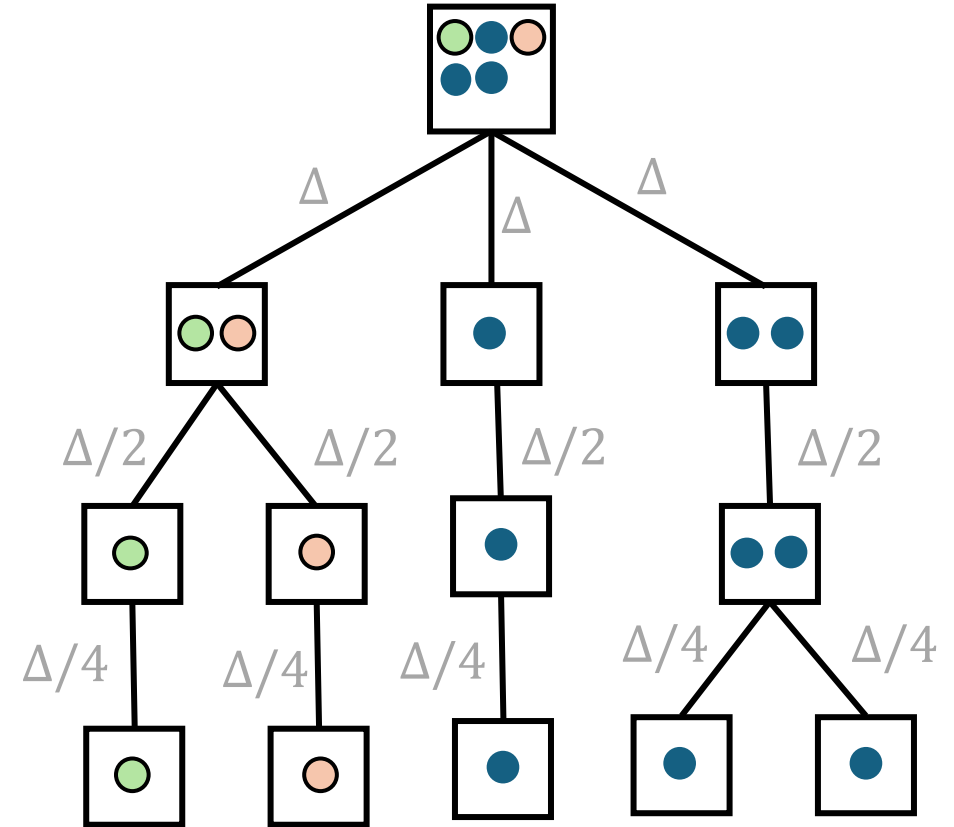


Distortion: Deterministic Lower Bound

- Lemma: For every $x, y \in X$ we have

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- Proof:

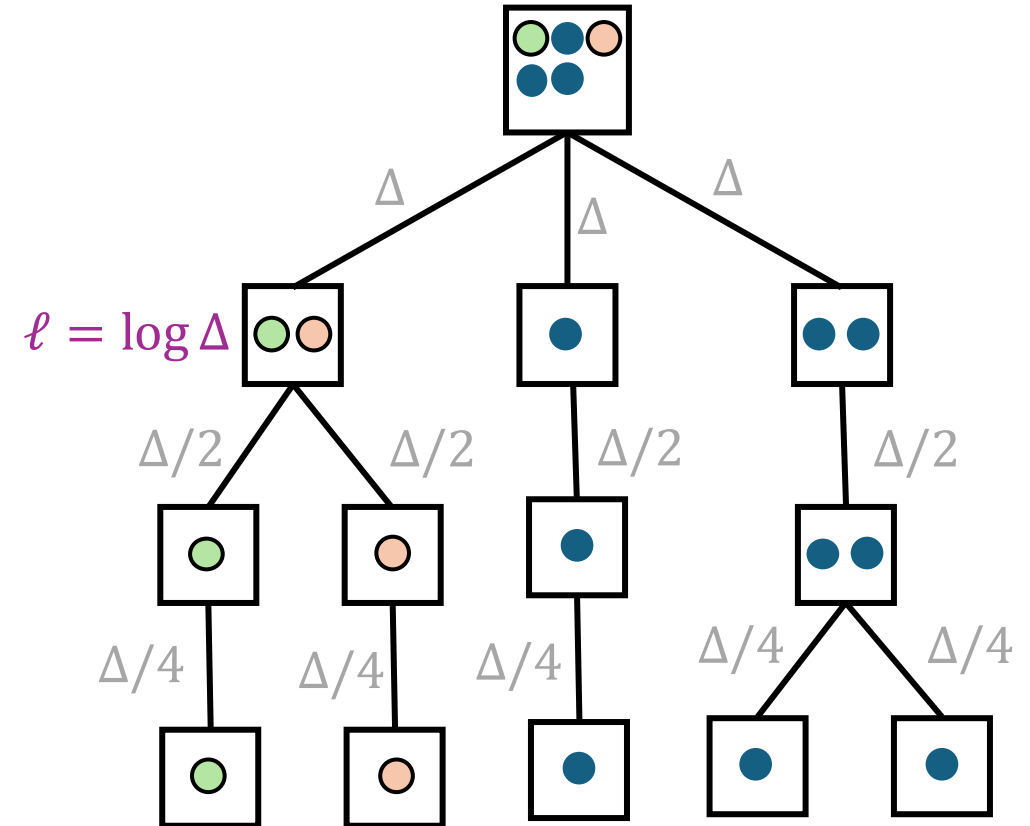


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 - Let ℓ be the lowest level where x, y are in the same node.

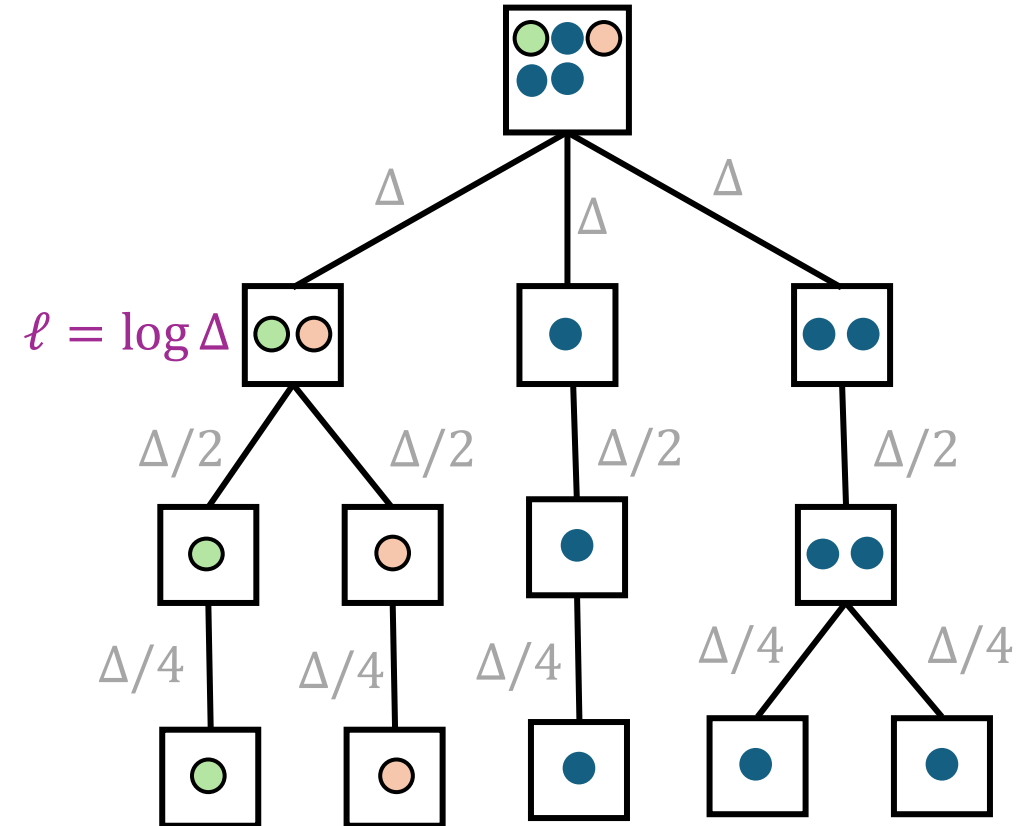


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- Lemma: For every $x, y \in X$ we have

$$\|x - y\|_1 \leq \textcolor{blue}{O}(d) \cdot T(x, y).$$

- Proof:
 - Let ℓ be the lowest level where x, y are in the same node.
 - Then, $T(x, y) \geq \Omega(2^\ell)$. Why?

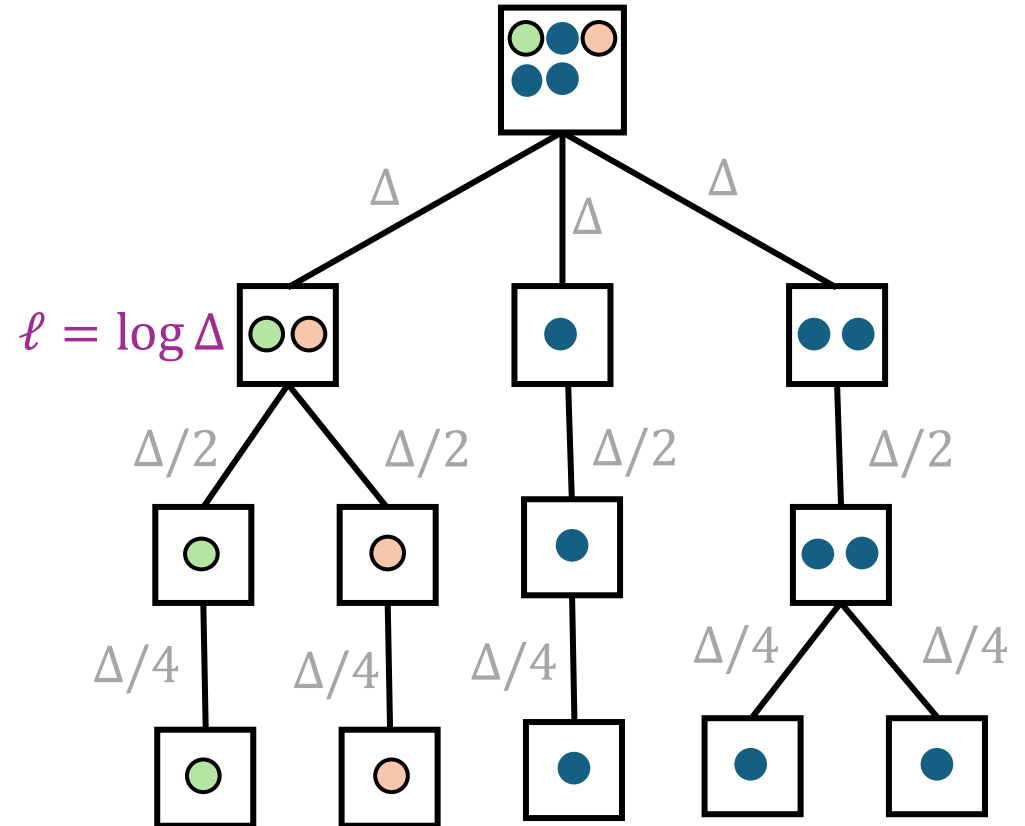


Distortion: Deterministic Lower Bound

- Lemma: For every $x, y \in X$ we have

$$\|x - y\|_1 \leq O(d) \cdot T(x, y).$$

- Proof:
 - Let ℓ be the lowest level where x, y are in the same node.
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 - On the other hand, each level- ℓ hypercube has side-length 2^ℓ , and hence ℓ_1 -diameter $2^\ell d$.



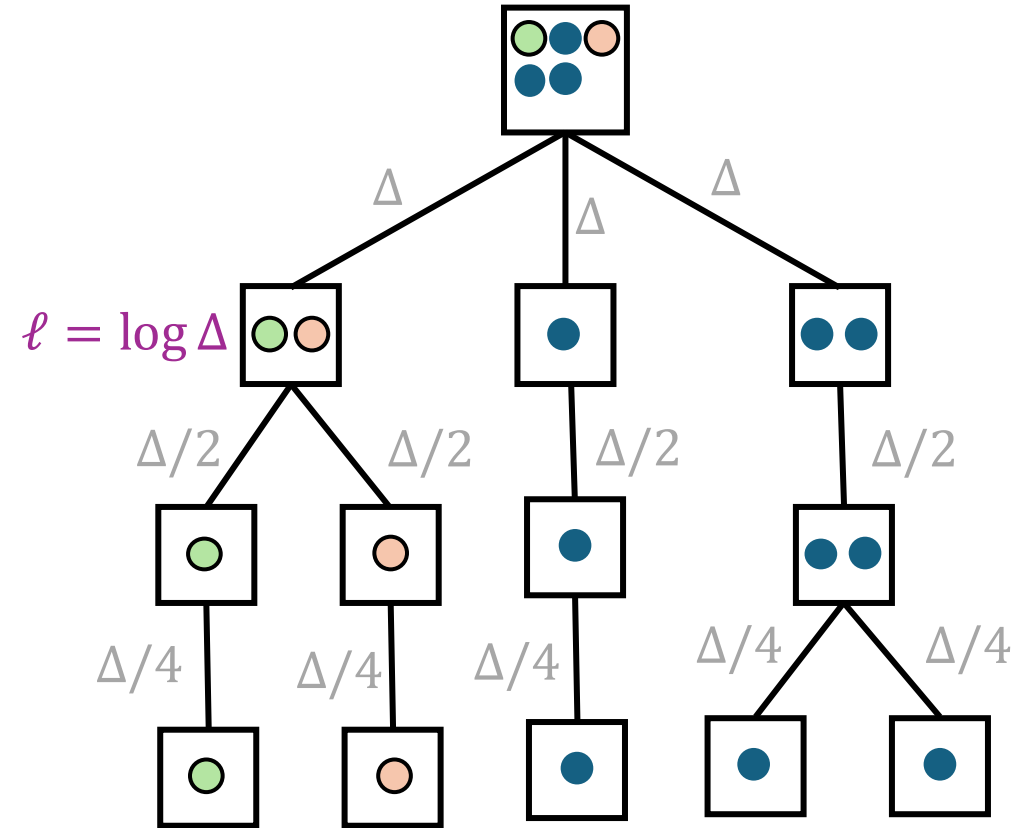
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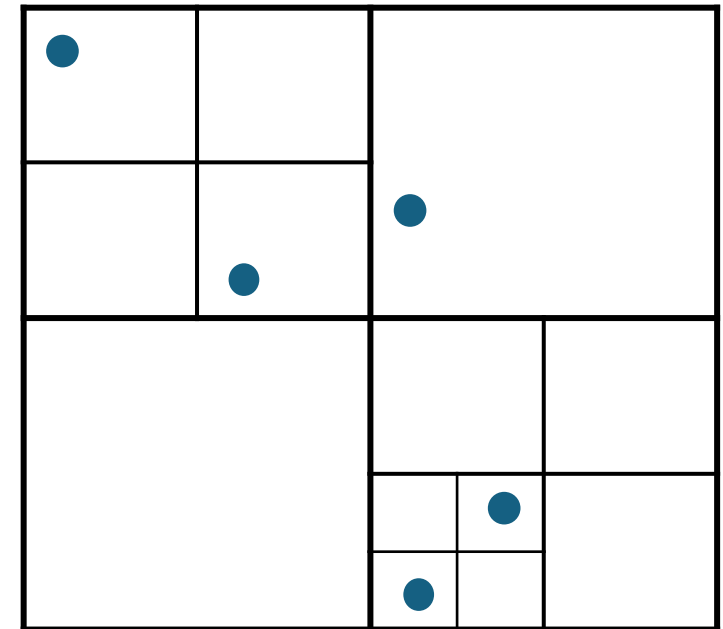
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- Therefore, $\|x - y\|_1 \leq 2^\ell d$.



Distortion: Expectation Upper Bound (1/2)

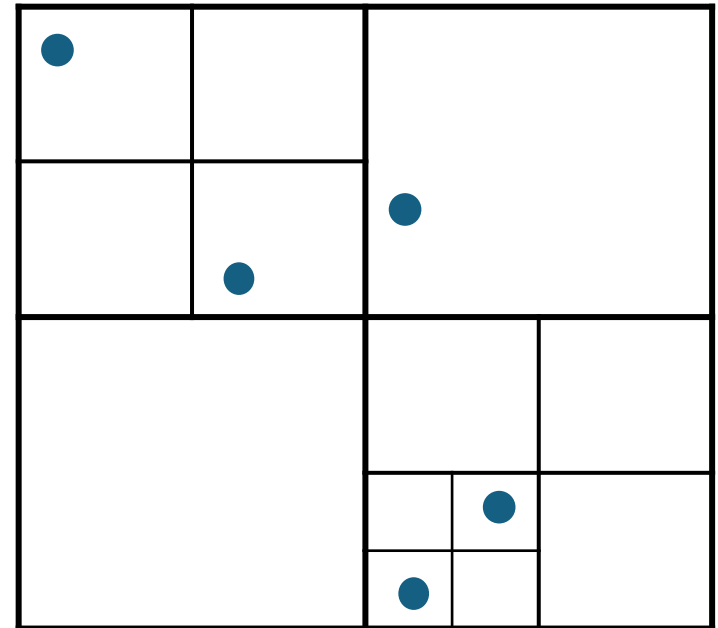
- Let $Sep(x, y, \ell)$ be the event that x, y are **not** in the same hypercube in level ℓ



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Why?



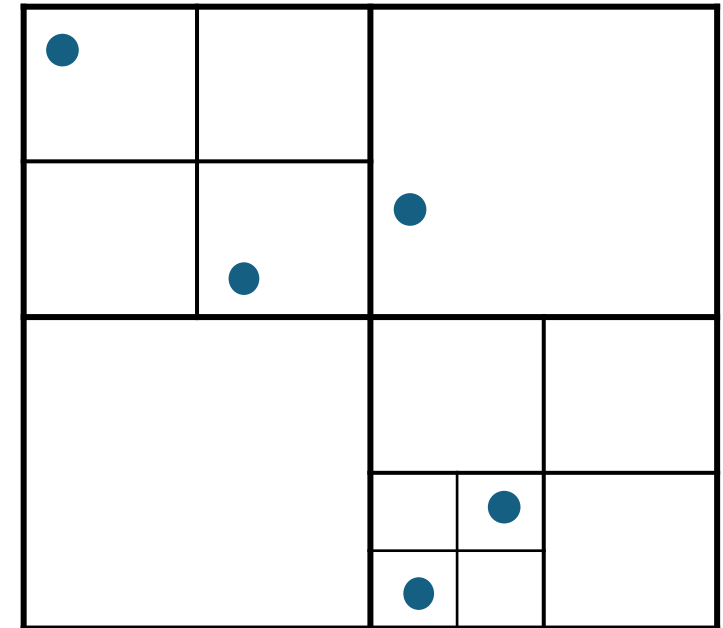
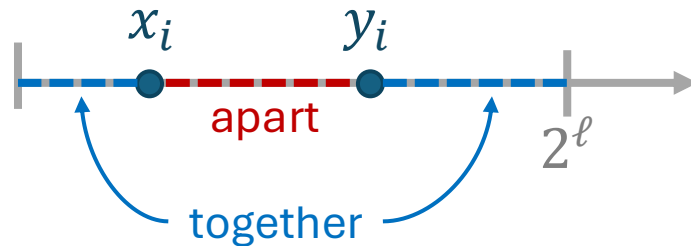
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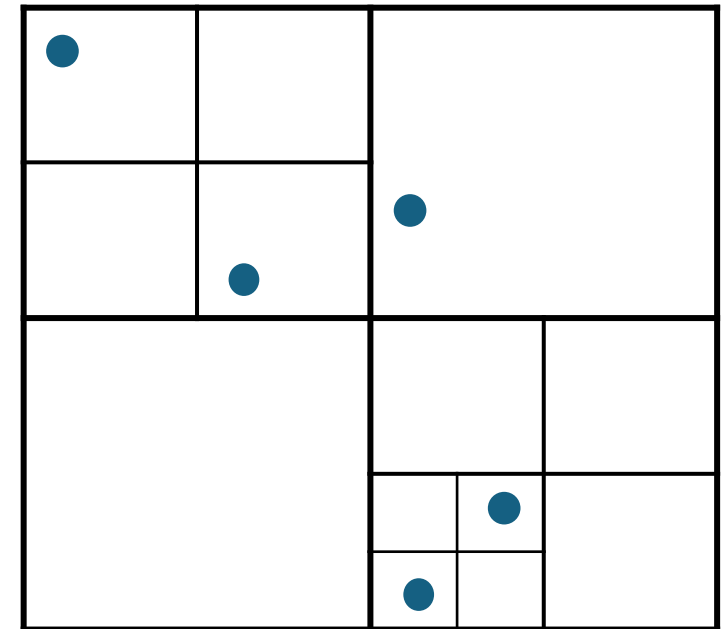
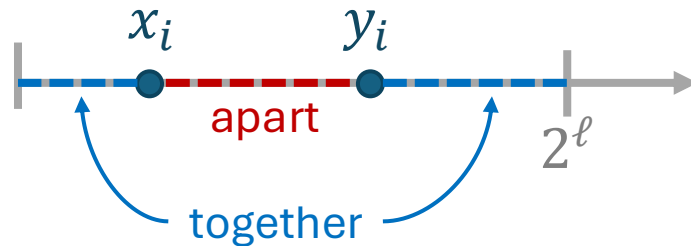
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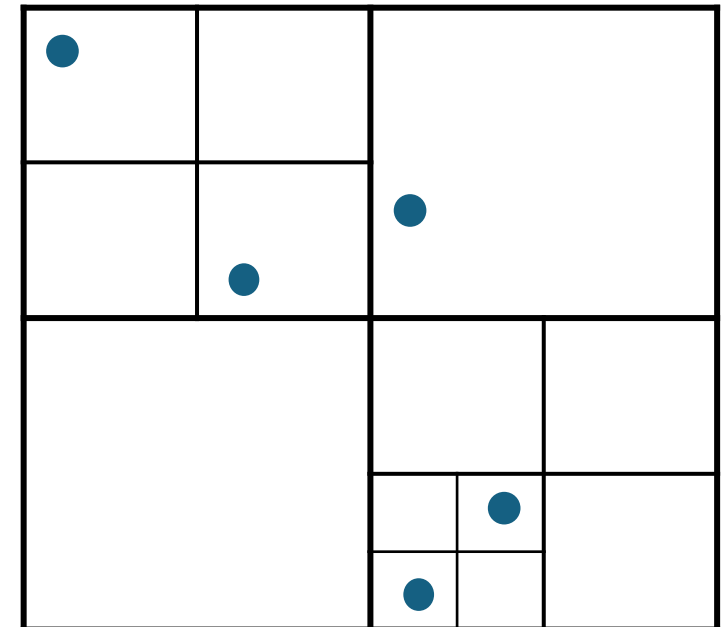
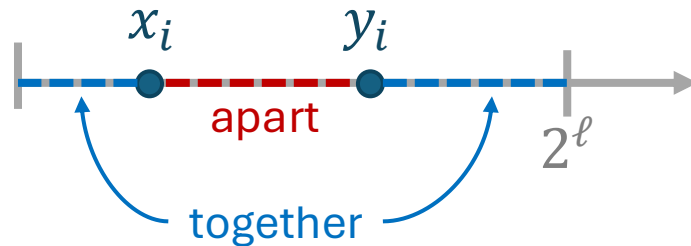
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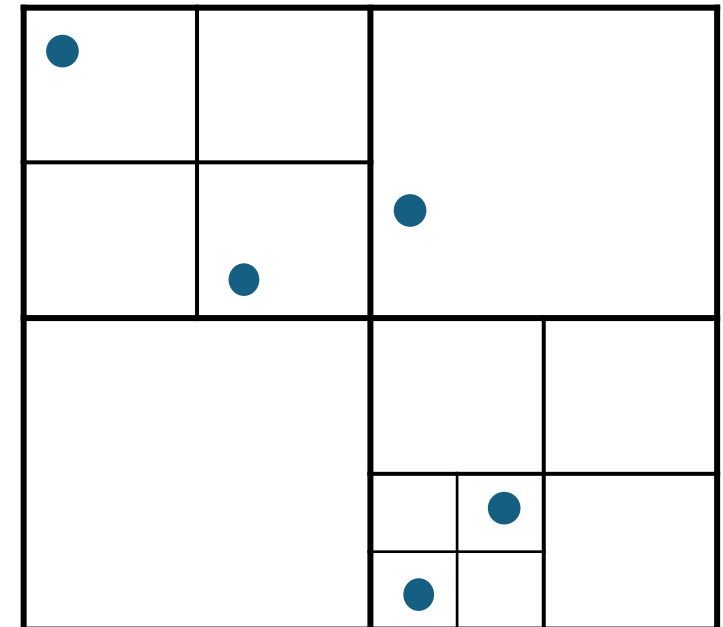
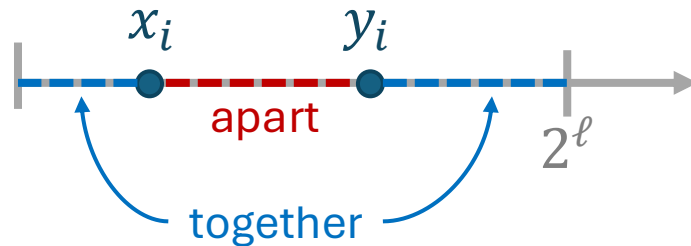


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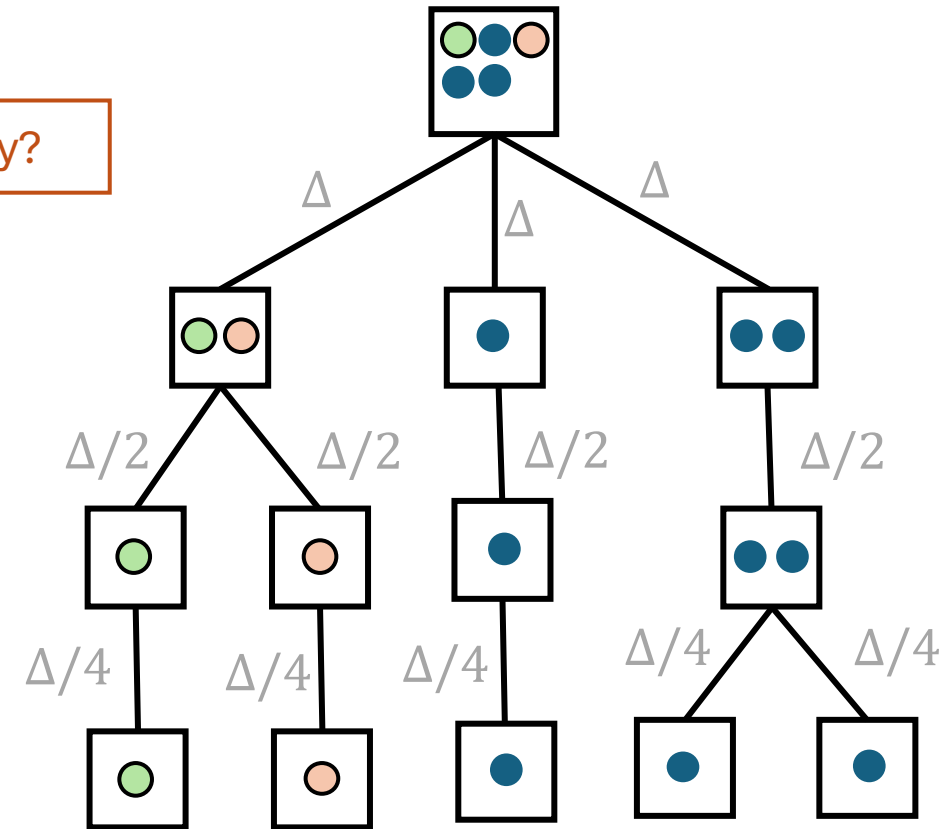


Distortion: Expectation Upper Bound (2/2)

- Observe:

$$T(x, y) = \sum_{\ell=\log(1/d)}^{\log(2\Delta)} \mathbf{1}\{Sep(x, y, \ell)\} \cdot 2 \cdot 2^\ell$$

Why?



Distortion: Expectation Upper Bound (2/2)

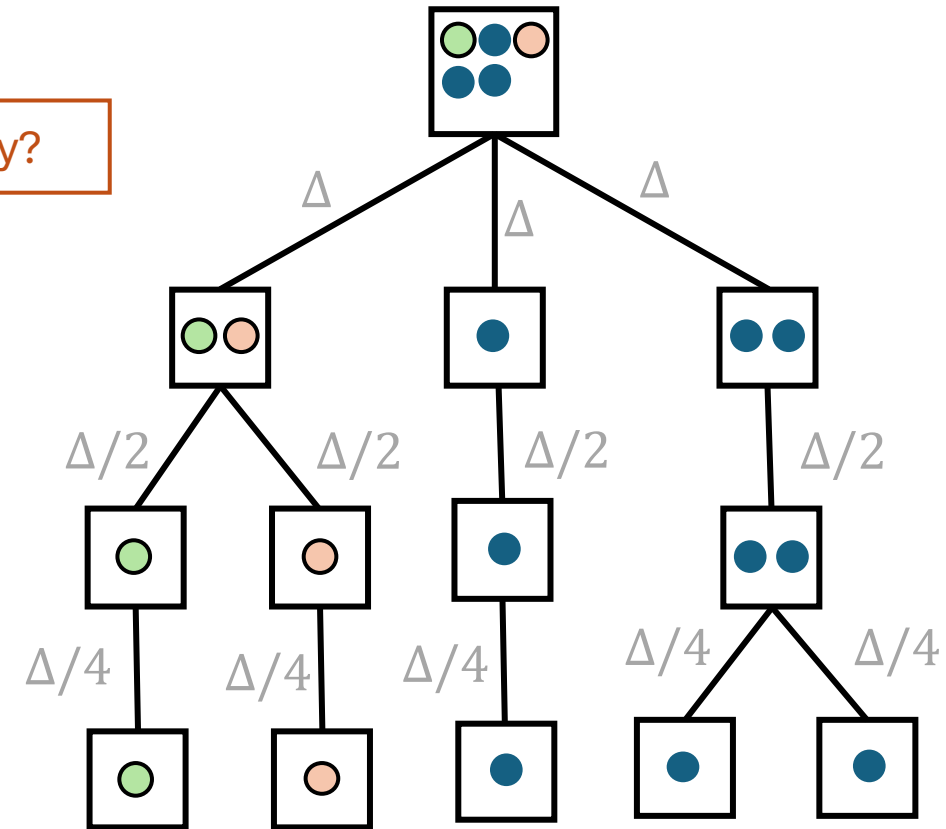
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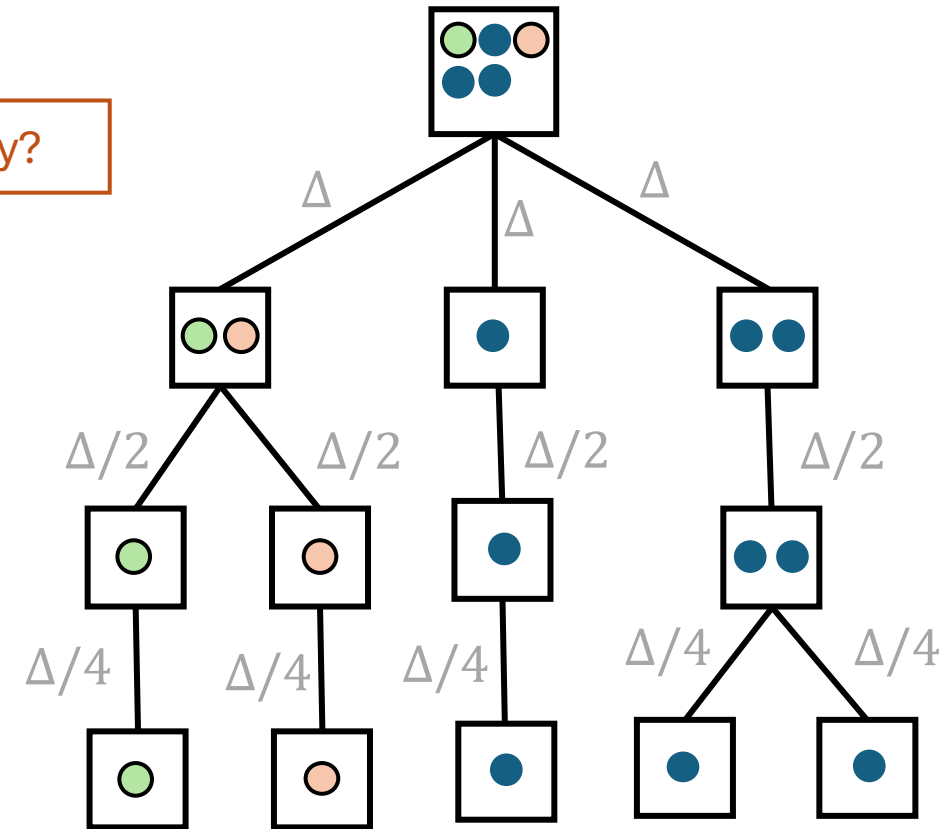
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$$\begin{aligned} \mathbb{E}[T(x, y)] &= \sum_{\ell=\log(1/d)}^{\log(2\Delta)} \Pr[Sep(x, y, \ell)] \cdot 2 \cdot 2^\ell \\ &\leq \sum_{\ell=\log(1/d)}^{\log(2\Delta)} 2^{-\ell} \|x - y\|_1 \cdot 2 \cdot 2^\ell \end{aligned}$$



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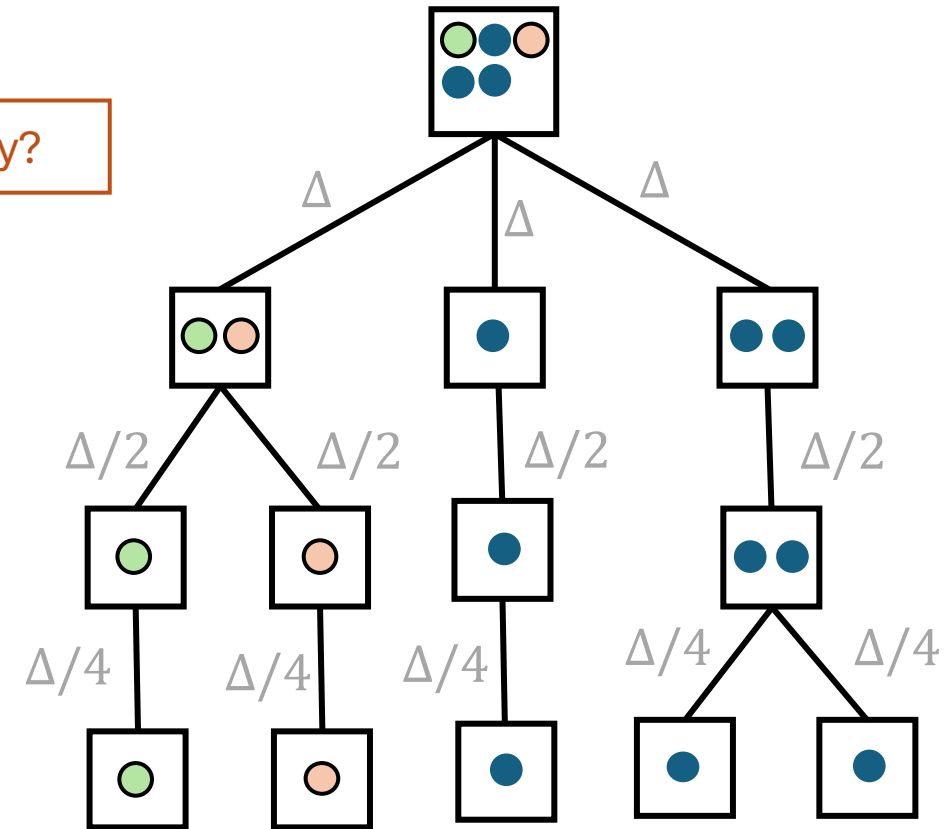
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$$\leq O(\log(\Delta d)) \cdot \|x - y\|_1$$

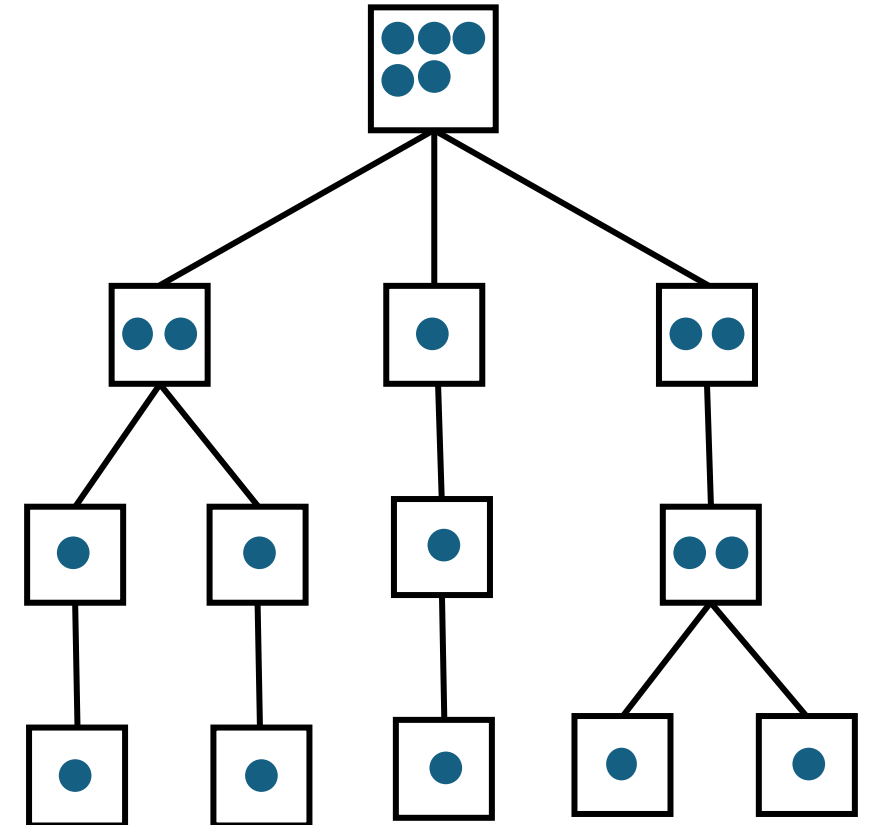


Approximate Nearest Neighbor Search

Faster but with larger approximation factor compared to last week

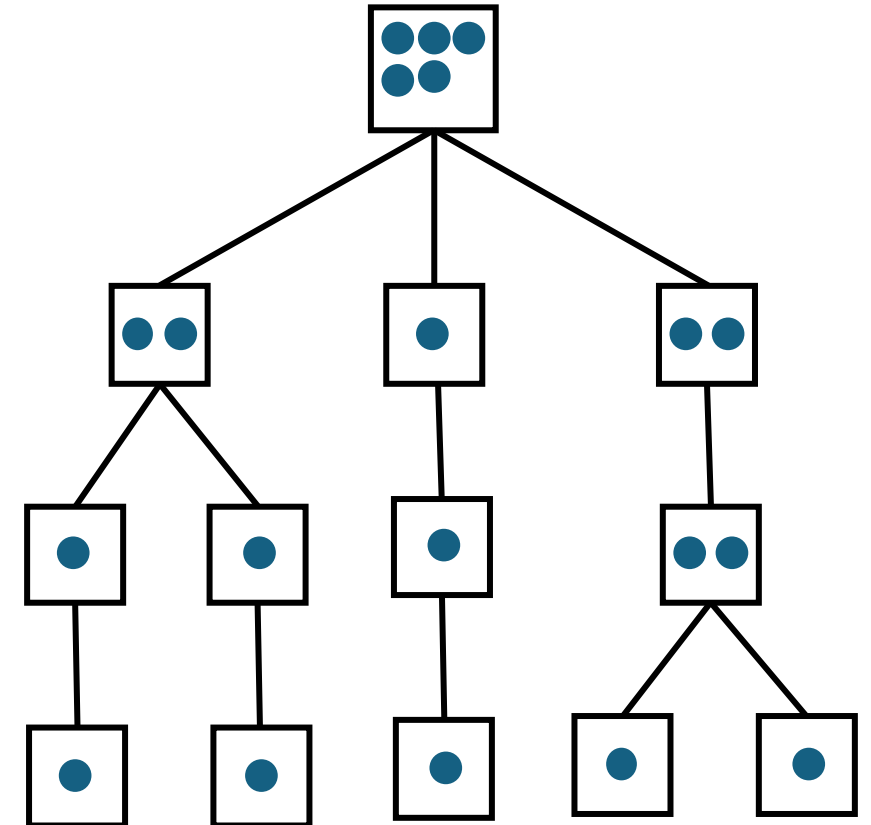
ANN with Quadrees

- **Theorem:** Quadtree yields c -approximate nearest neighbor search with:
 - Construction time and memory: $O(nd \log(\Delta d)) \approx n \log^{O(1)} n$
 - Query time: $O(d \log(\Delta d)) \approx \log^{O(1)} n$
 - Approximation: $c = O(d^2 \log(\Delta d)) \approx \log^{O(1)} n$



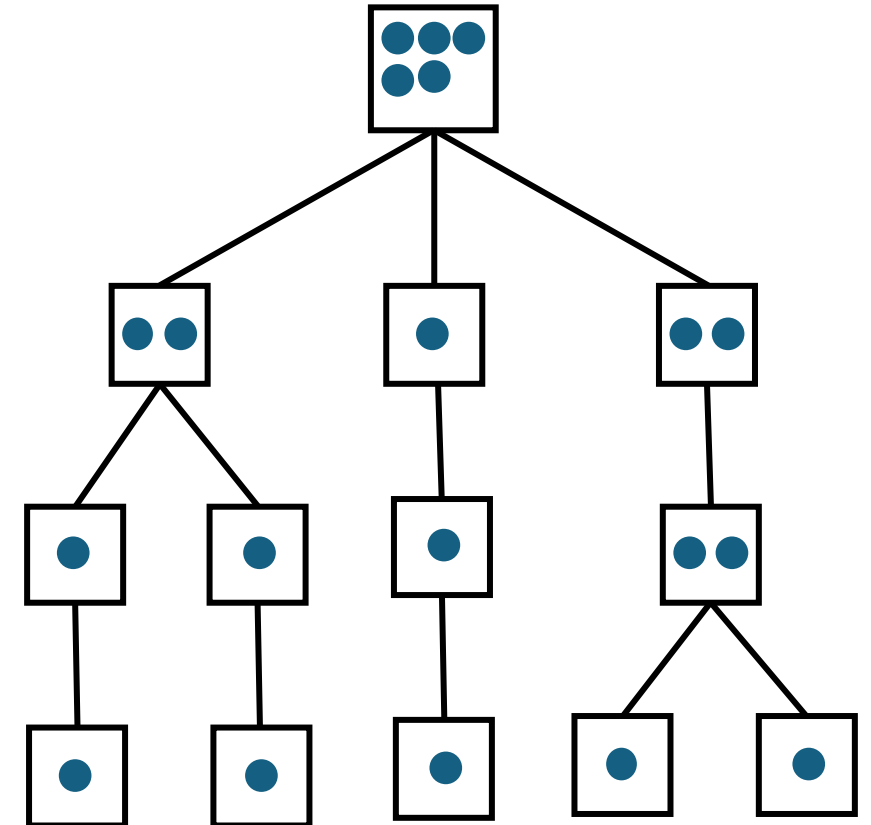
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- **Construction:** Build quadtree on dataset
 - Why is the memory $O(nd \log(\Delta d))$?



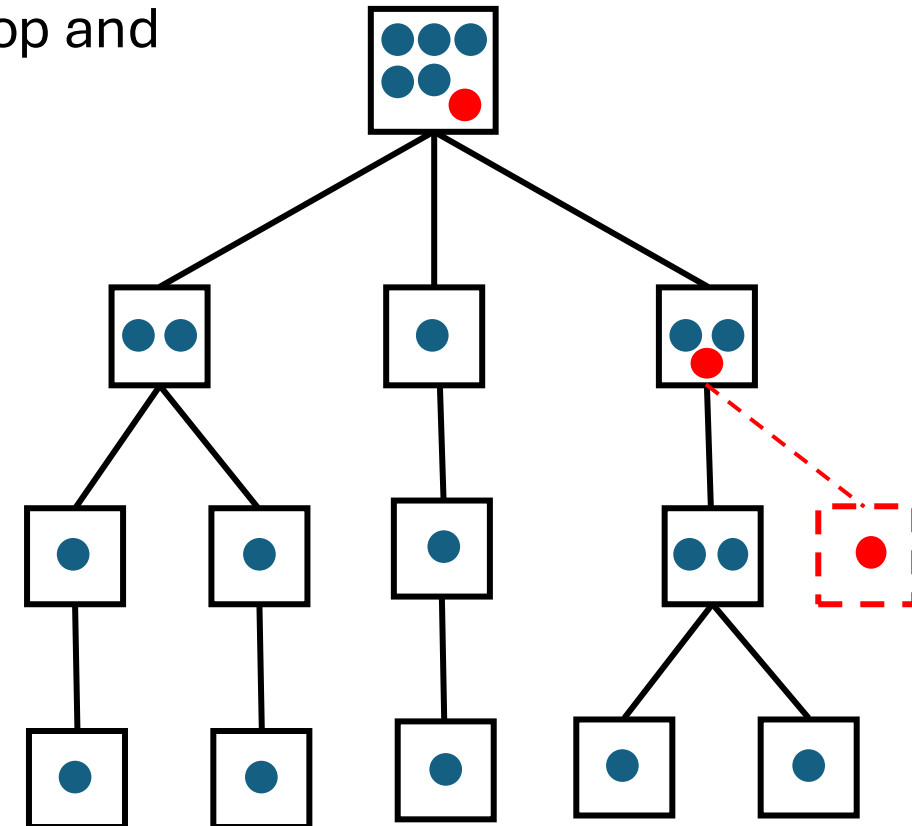
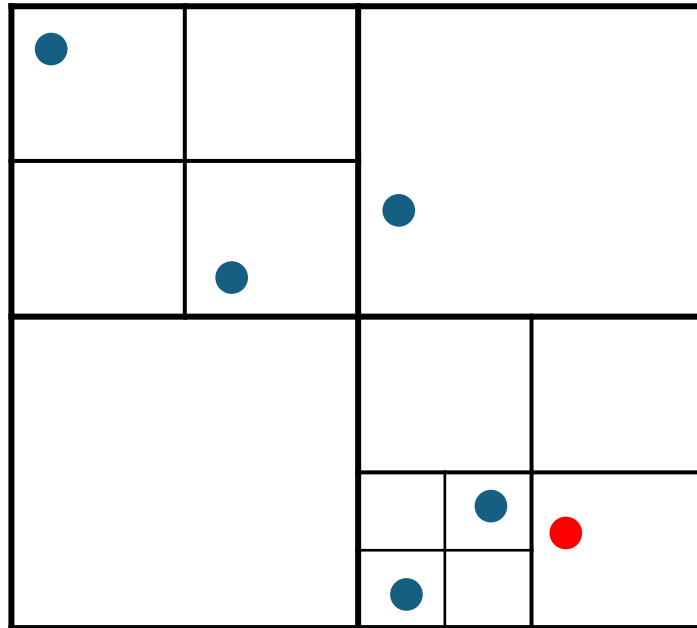
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- **Query algorithm:** ?
 - The query point is not in the tree
 - Its hypercubes in some levels might not exist



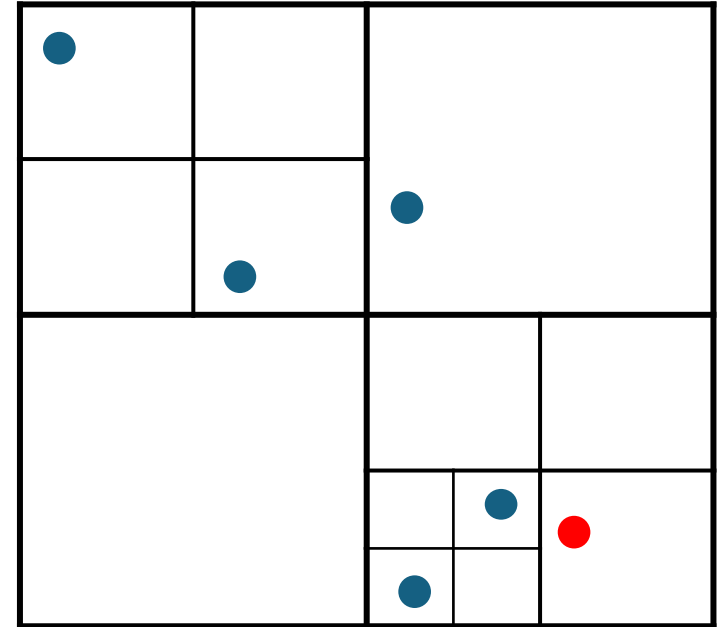
ANN with Quadtrees: Query

- Start at the root
- Traverse downward as if inserting the query point into the tree
- If the corresponding downward edge does not exist, stop and return any point in the current hypercube



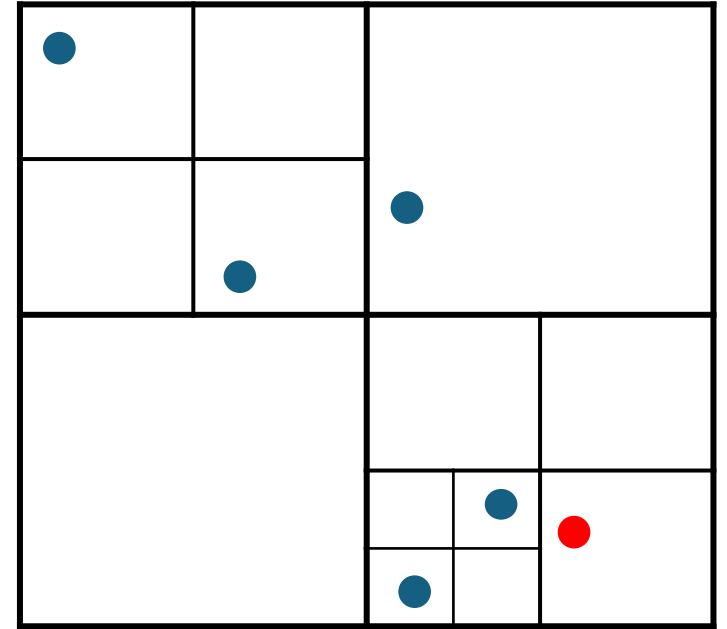
ANN with Quadtrees: Analysis

- Notation:
 - Let y be the query point
 - Let x^* be the true nearest neighbor
 - Let $\tilde{x}$ be the returned nearest neighbor



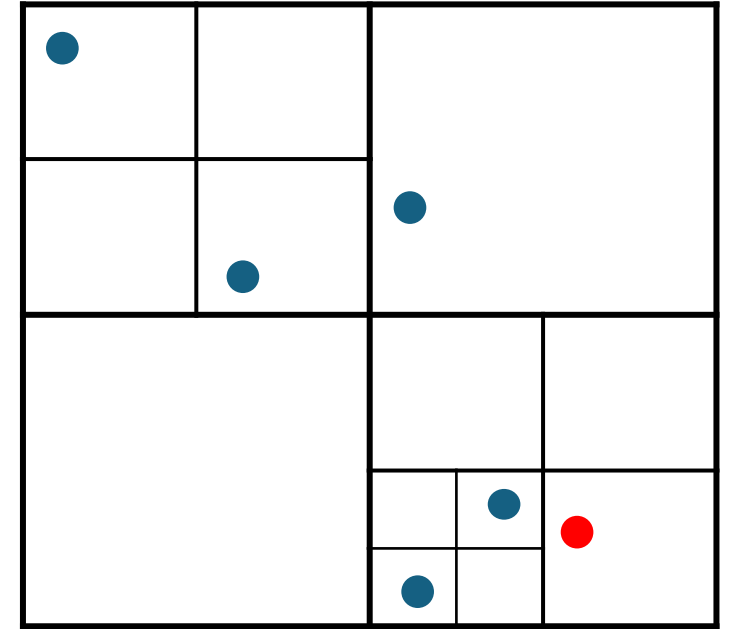
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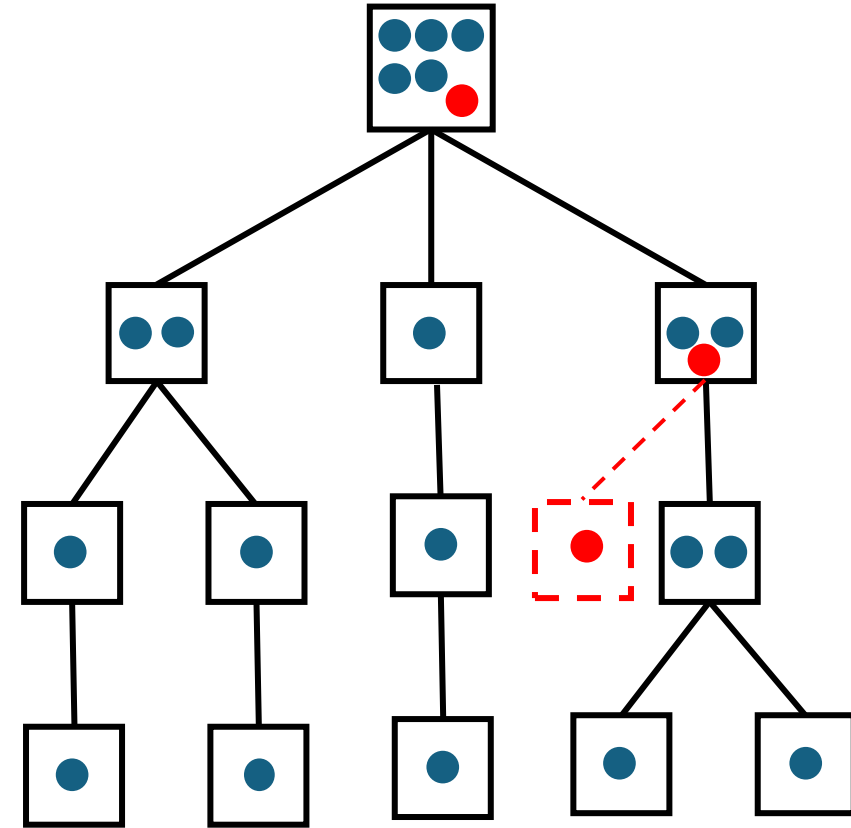
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 - *Intuitively*: If y, x^* are “near” at distance scale $2^{-\ell} \Delta$, then they collide in the random partition induced by the level- ℓ grid
 - $\alpha \in (0,1)$, and we want it to be as large as possible



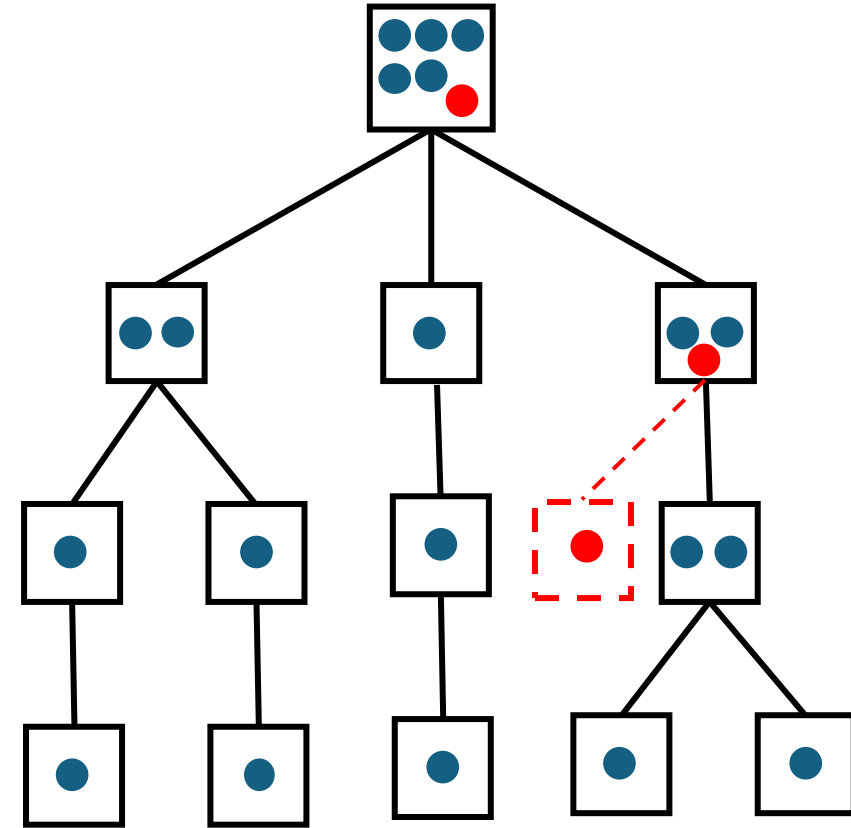
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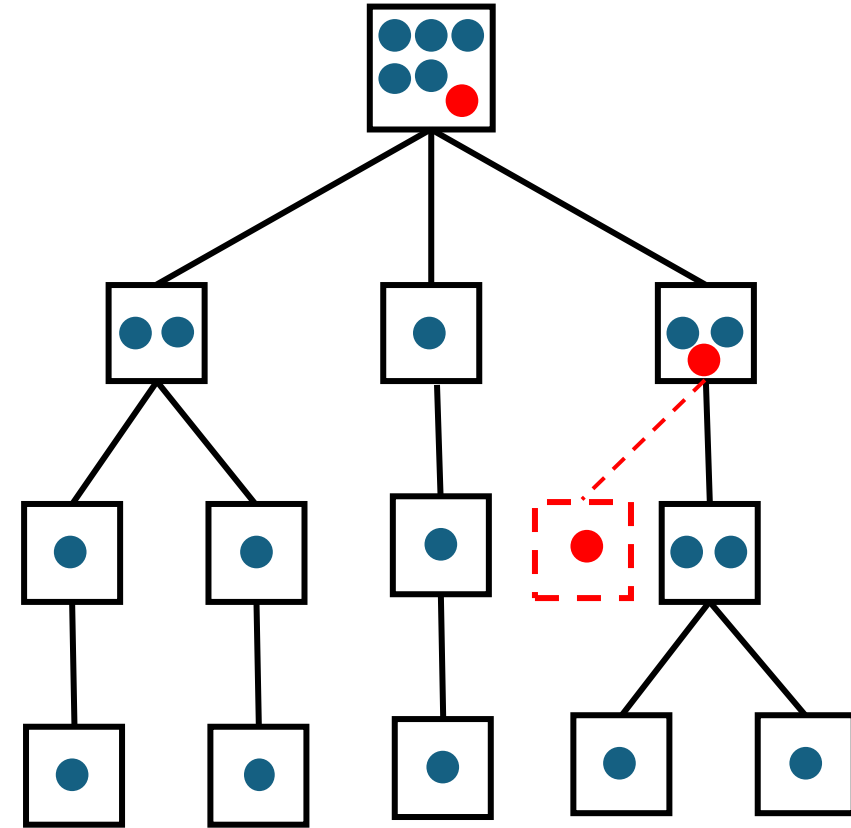
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 - In other words: the quadtree reports a $(2d\alpha^{-1})$ -approximate nearest neighbor



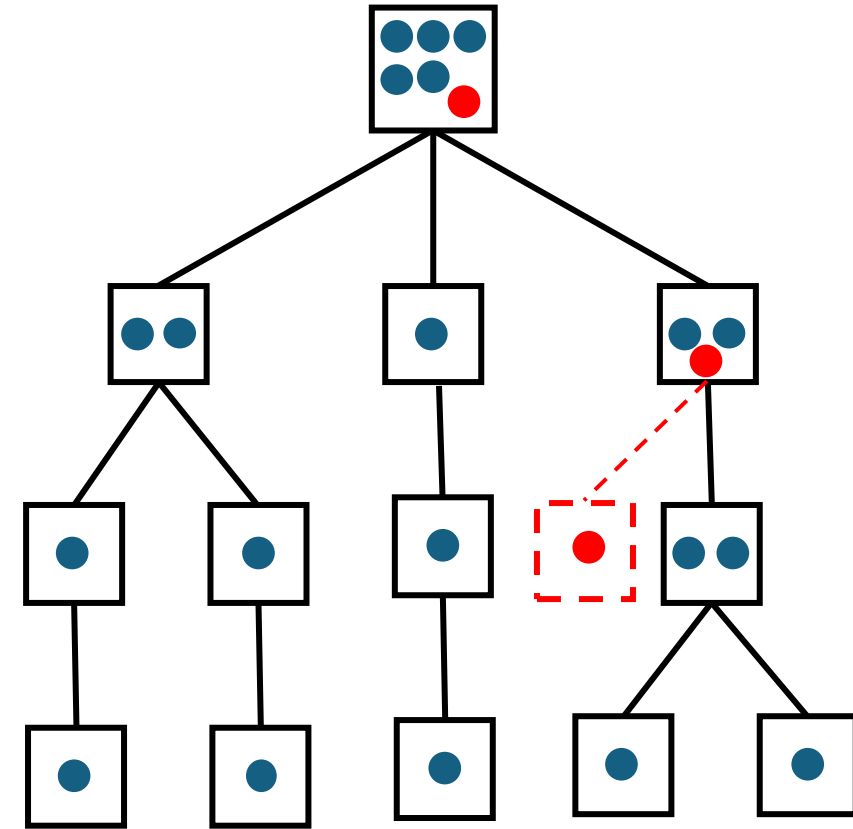
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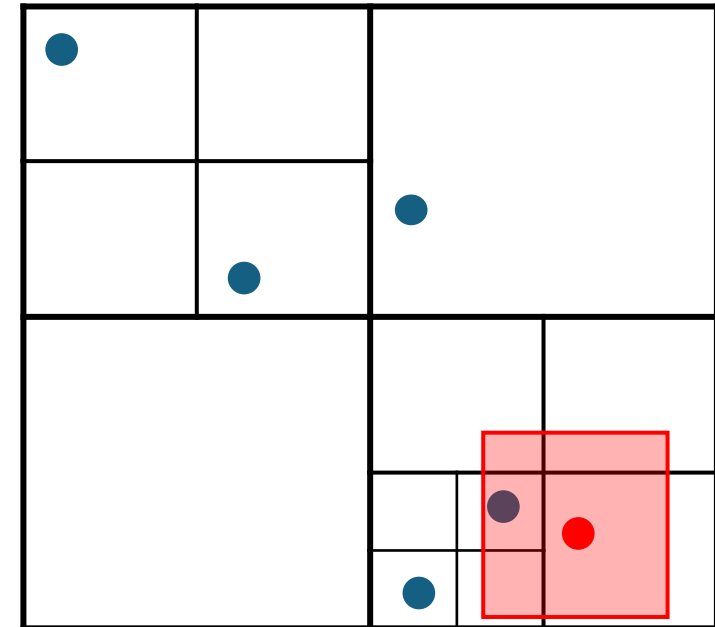
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- Proof:
 - Let ℓ^* be the lowest level where y, x^* are in the same hypercube.
 - In level $\ell^* + 1$ they are separated, hence y α -goodness: $\|y - x^*\|_1 \geq \alpha \cdot 2^{-\ell^*-1} \Delta$.
 - The query algorithm returns some point $\tilde{x}$ from the same level- ℓ^* hypercube as y , hence: $\|y - \tilde{x}\|_1 \leq 2^{-\ell^*} \Delta$.



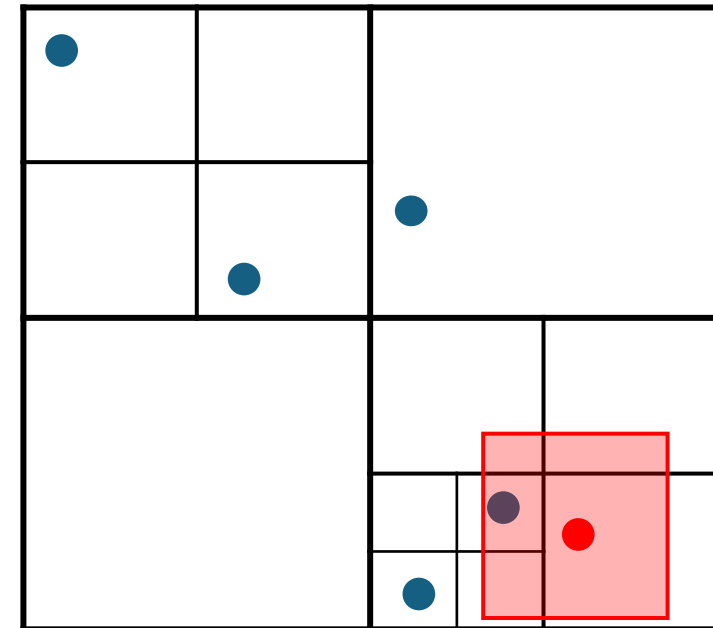
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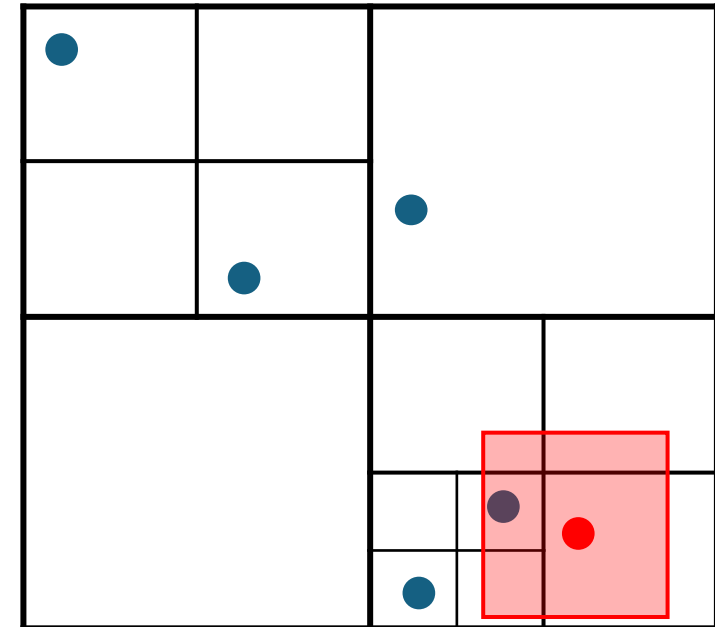
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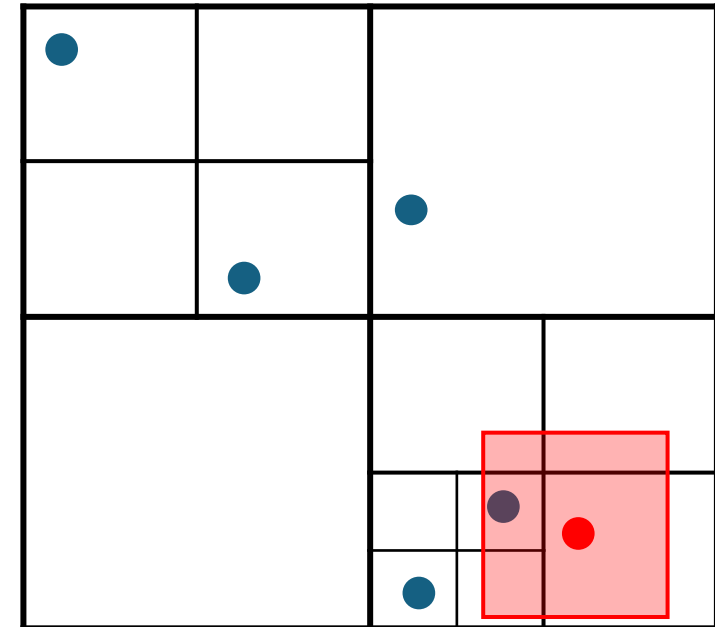
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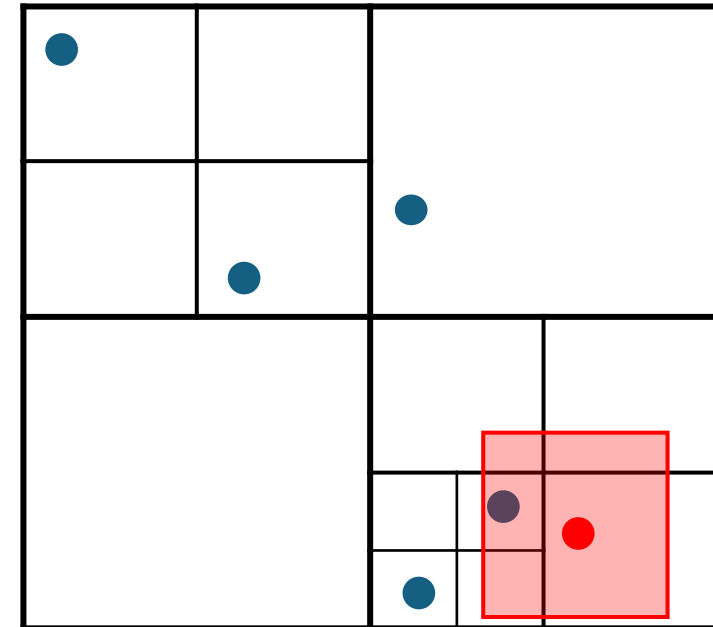
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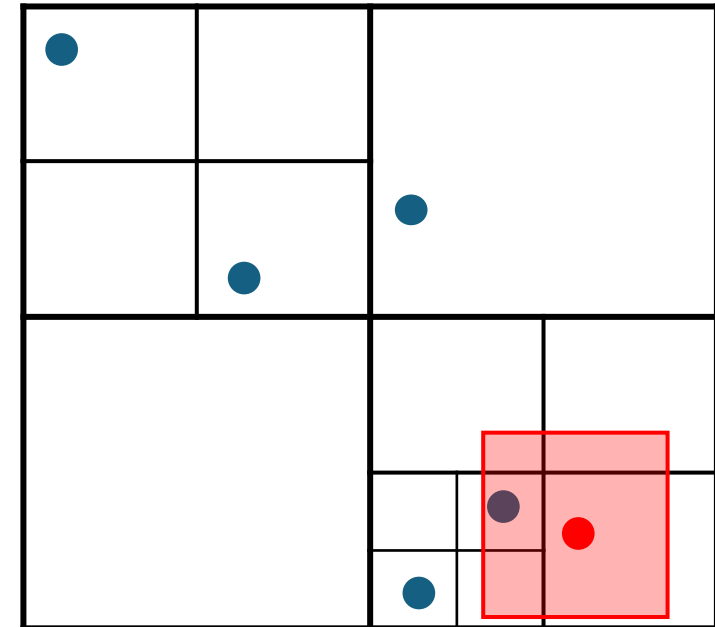
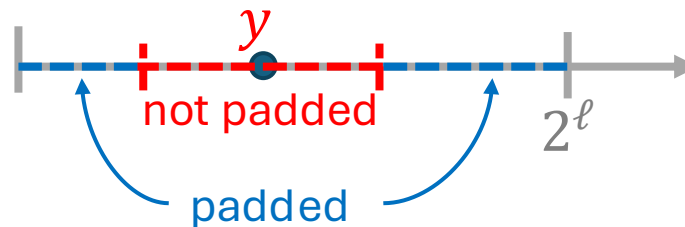
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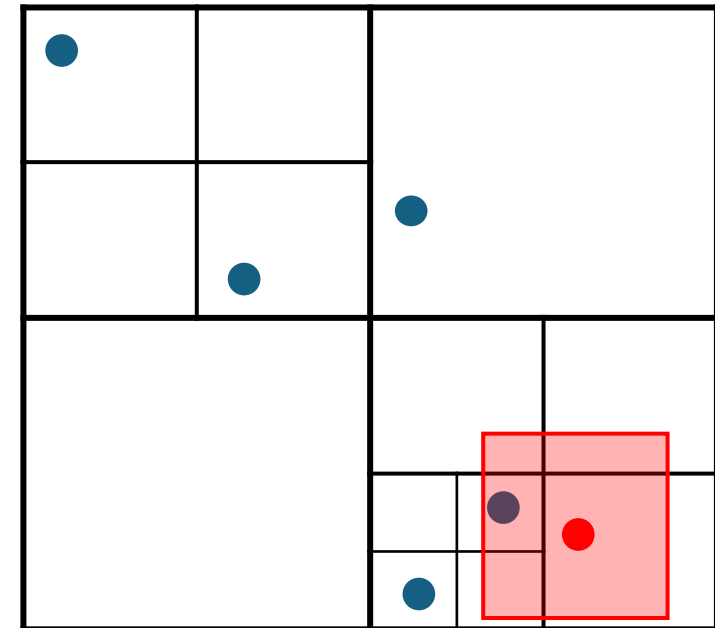
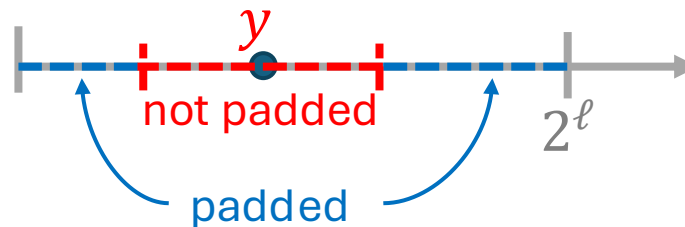
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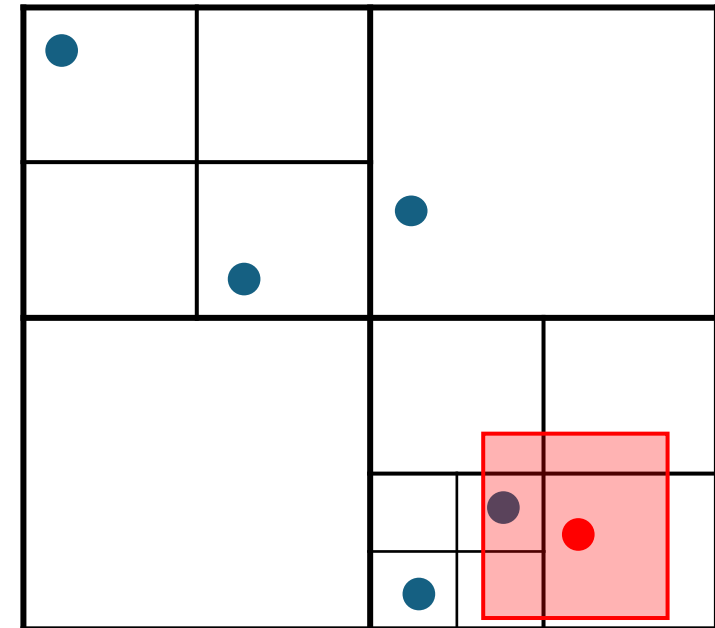
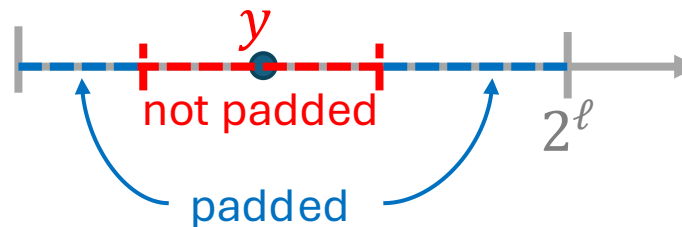
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- ANN theorem proven.

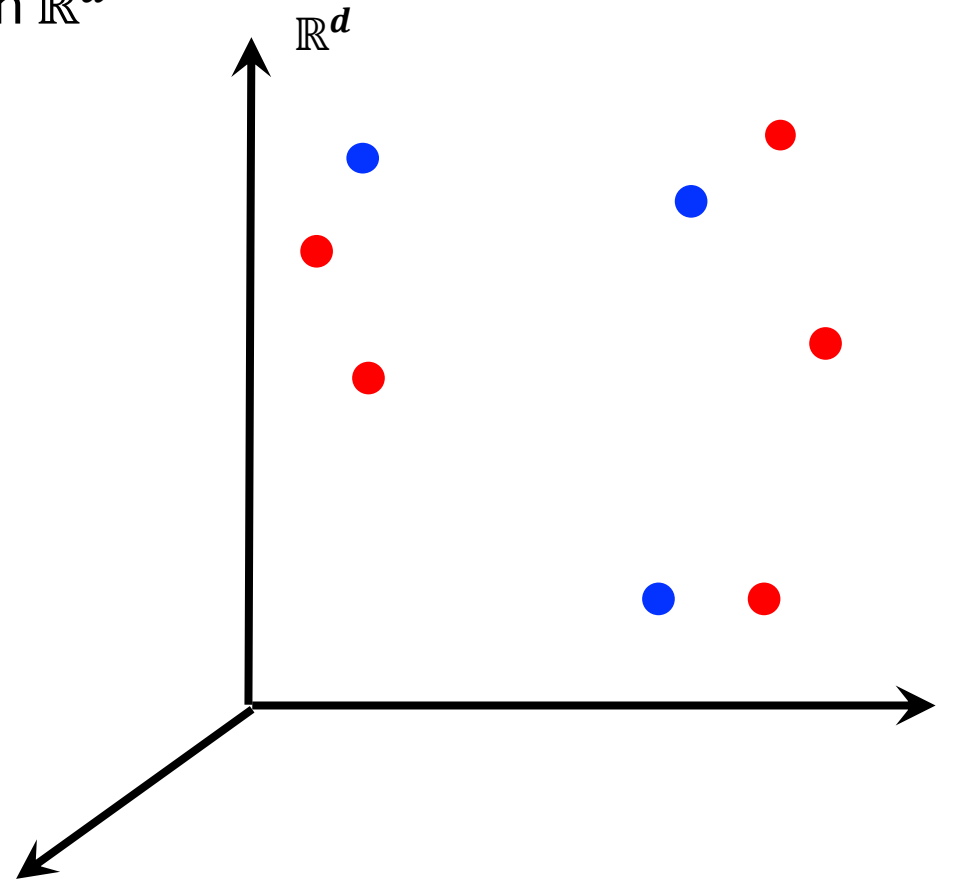


Earth Mover Distance (EMD)

A.k.a, Optimal Transport, Wasserstein-1

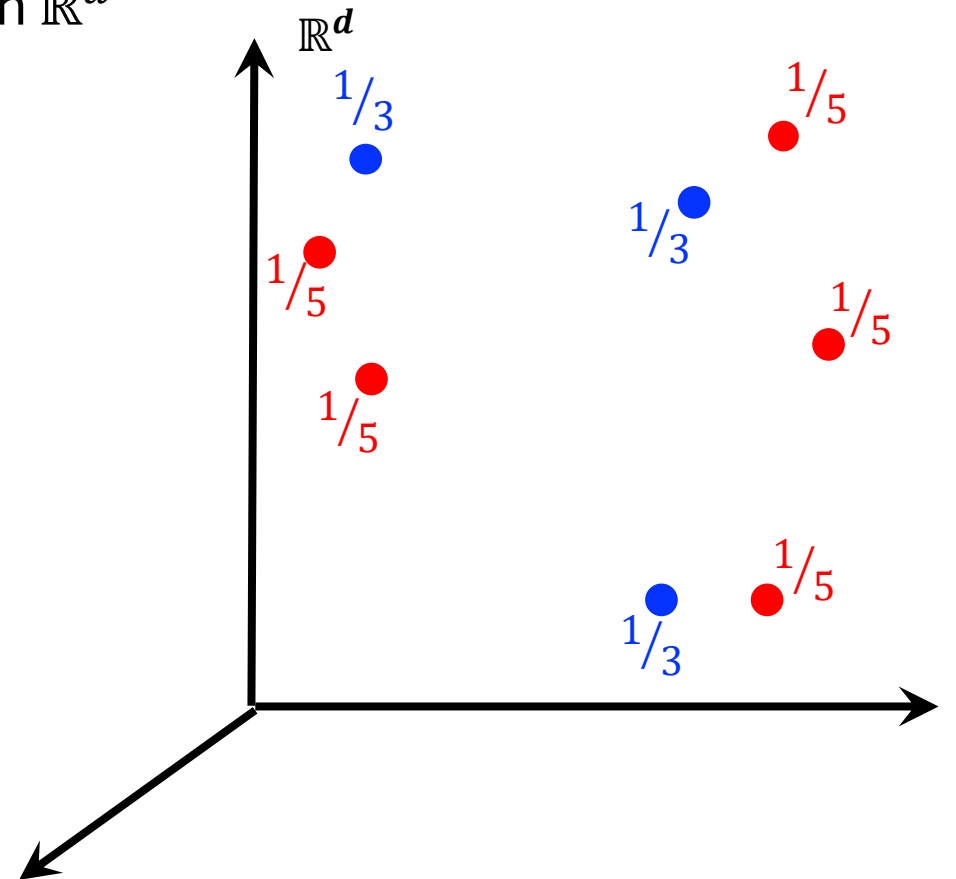
Earth Mover Distance

- EMD is a metric between distributions over points in $\mathbb{R}^d$



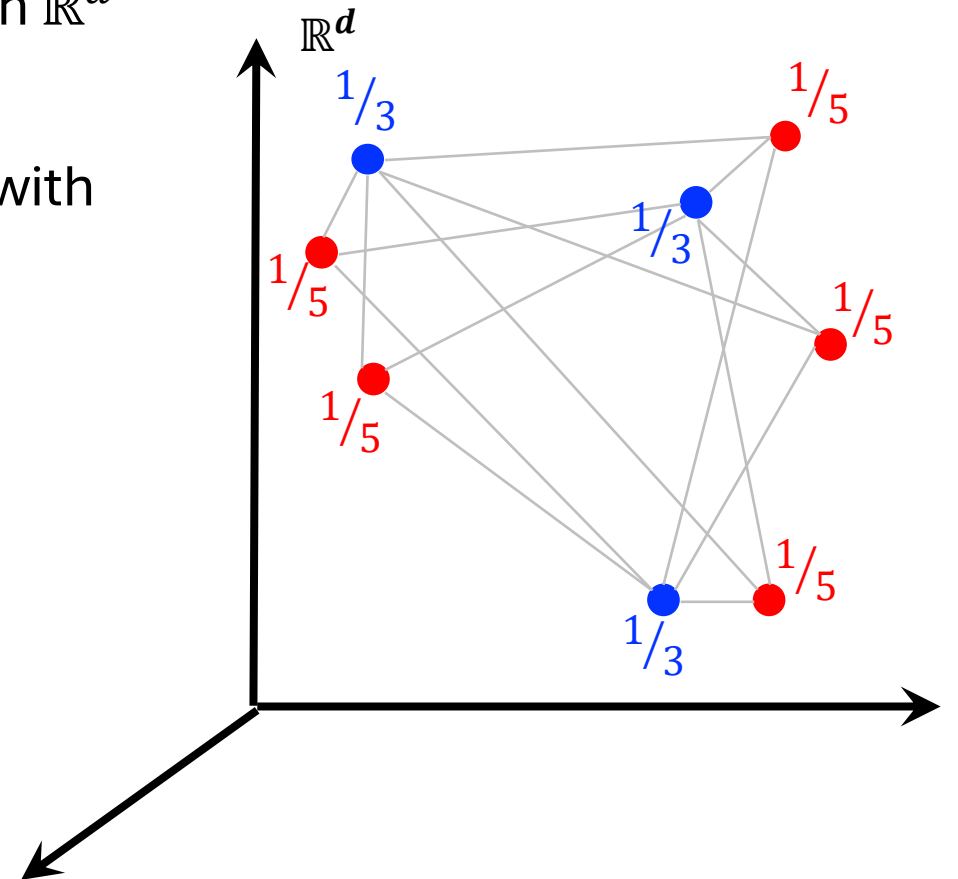
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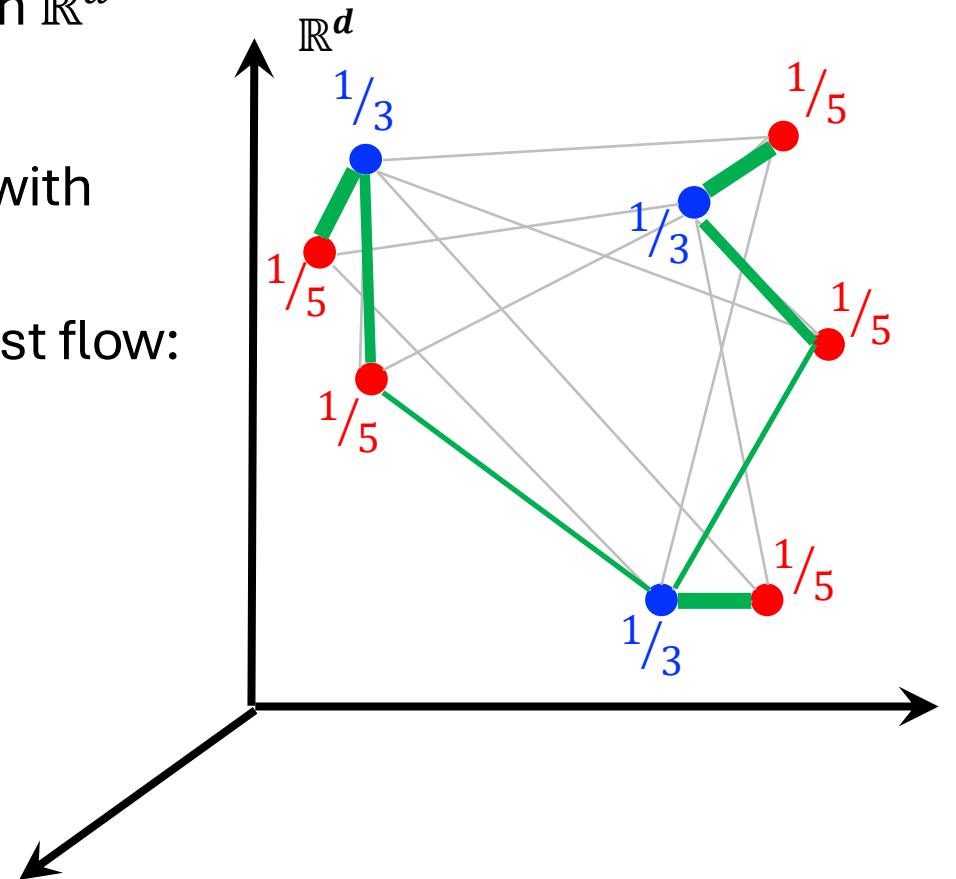
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- Consider the bipartite graph between their points, with edge weights equal to Euclidean (or ℓ_1) distances
- The EMD between X and Y is defined as the min-cost flow:

$$EMD(X, Y) = \min_{\mathbf{F}} \sum_{x \in X, y \in Y} \|x - y\| \cdot \mathbf{F}(x, y)$$

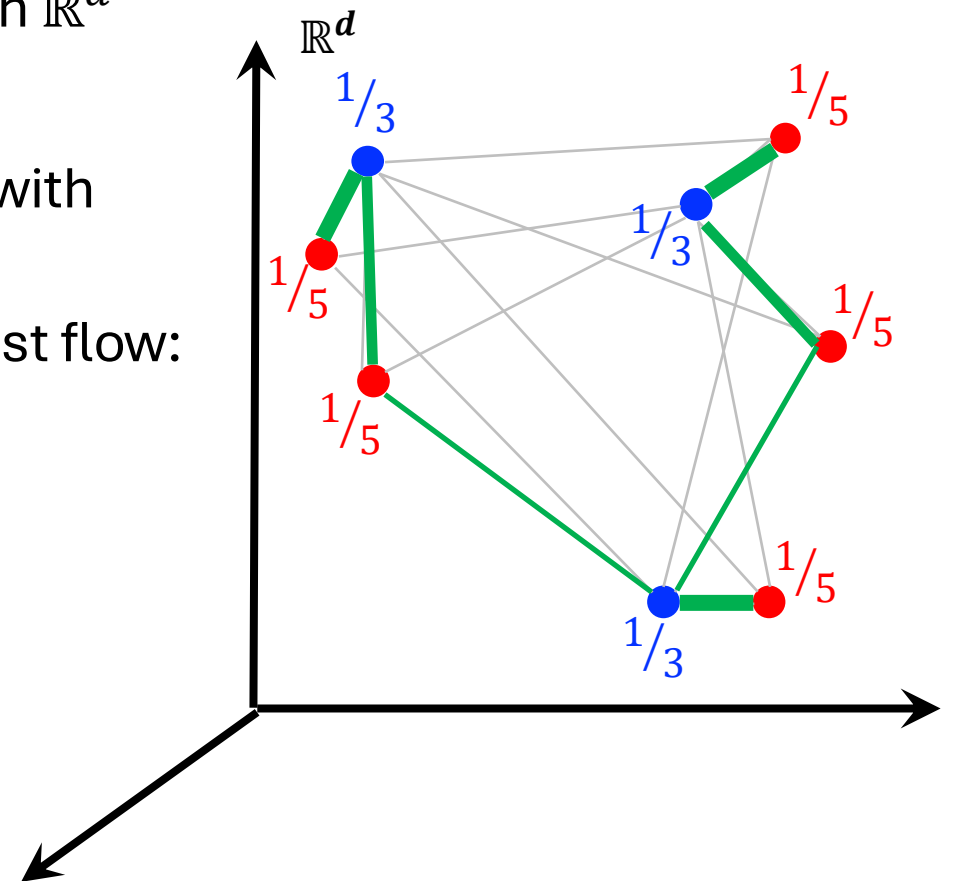


Earth Mover Distance

- EMD is a metric between distributions over points in $\mathbb{R}^d$
- Let X and Y be two such distributions
- Consider the bipartite graph between their points, with edge weights equal to Euclidean (or ℓ_1) distances
- The EMD between X and Y is defined as the min-cost flow:

$$EMD(X, Y) = \min_{\mathbf{F}} \sum_{x \in X, y \in Y} \|x - y\| \cdot \mathbf{F}(x, y)$$

- How much time does it take to compute exactly?



Application: “Word Mover Distance”

Kusner et al. (2015): Use OT as distance between text documents

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Document 1

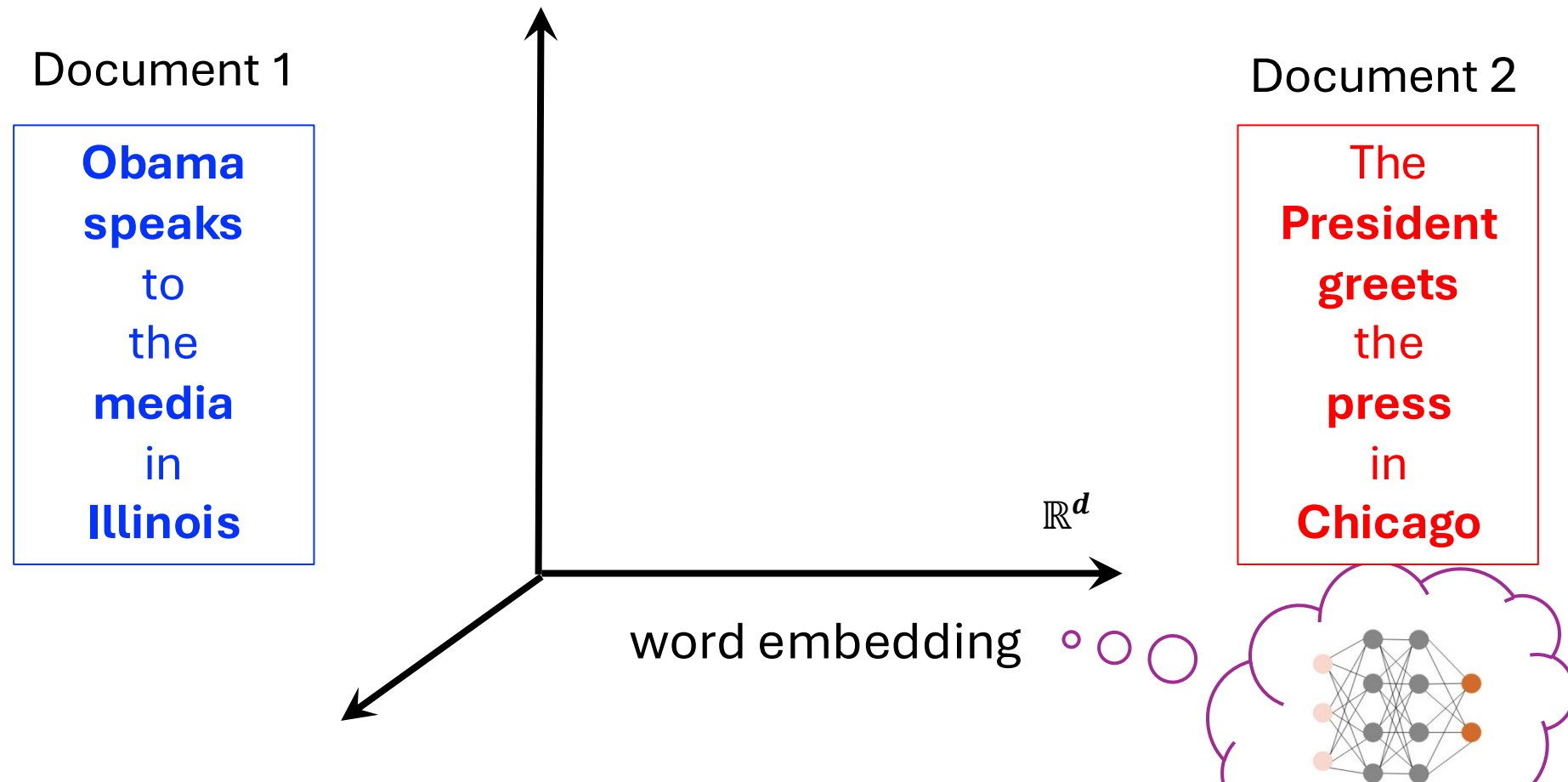
Obama
speaks
to
the
media
in
Illinois

Document 2

The
President
greets
the
press
in
Chicago

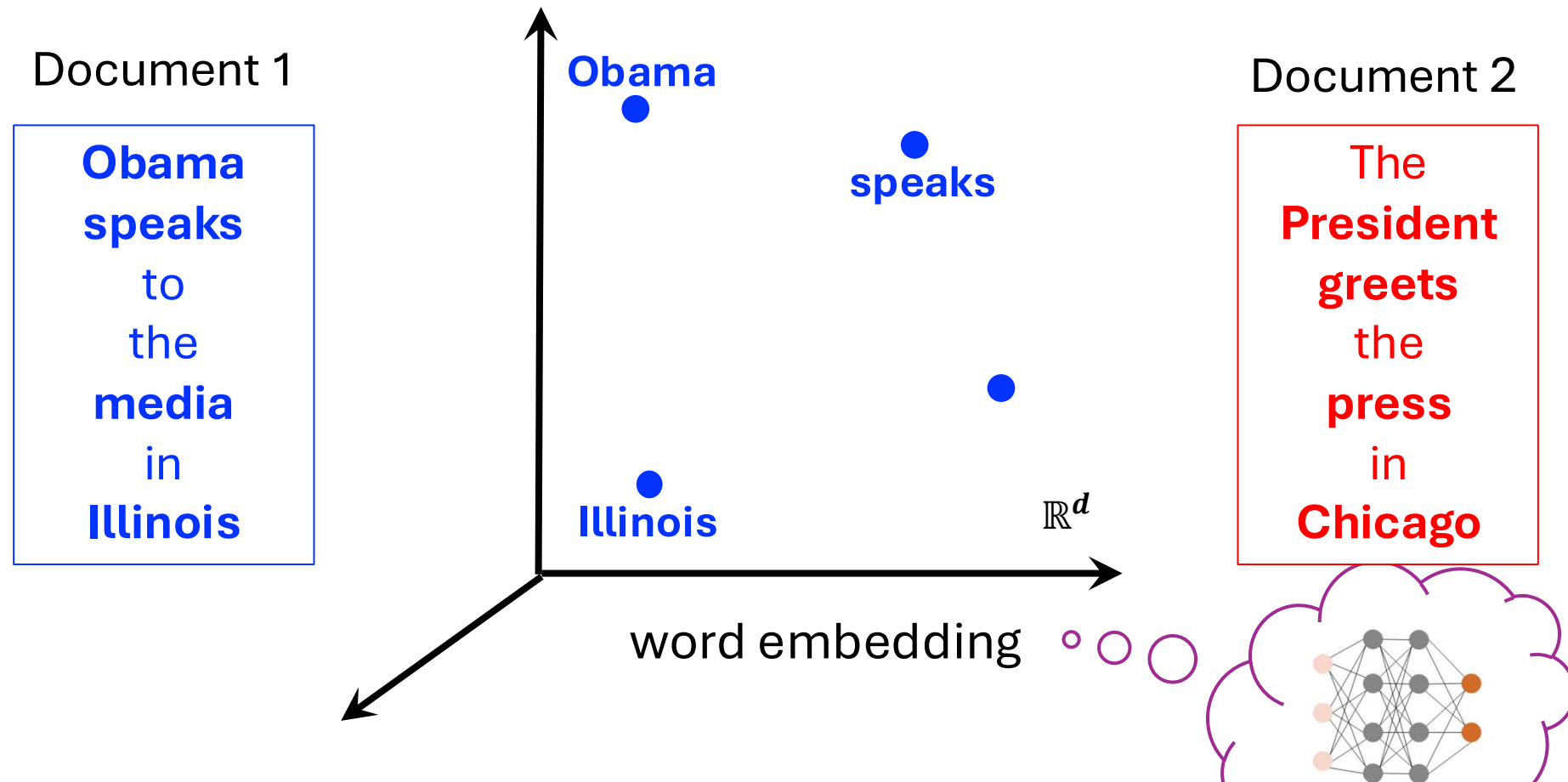
Application: “Word Mover Distance”

Kusner et al. (2015): Use OT as distance between text documents



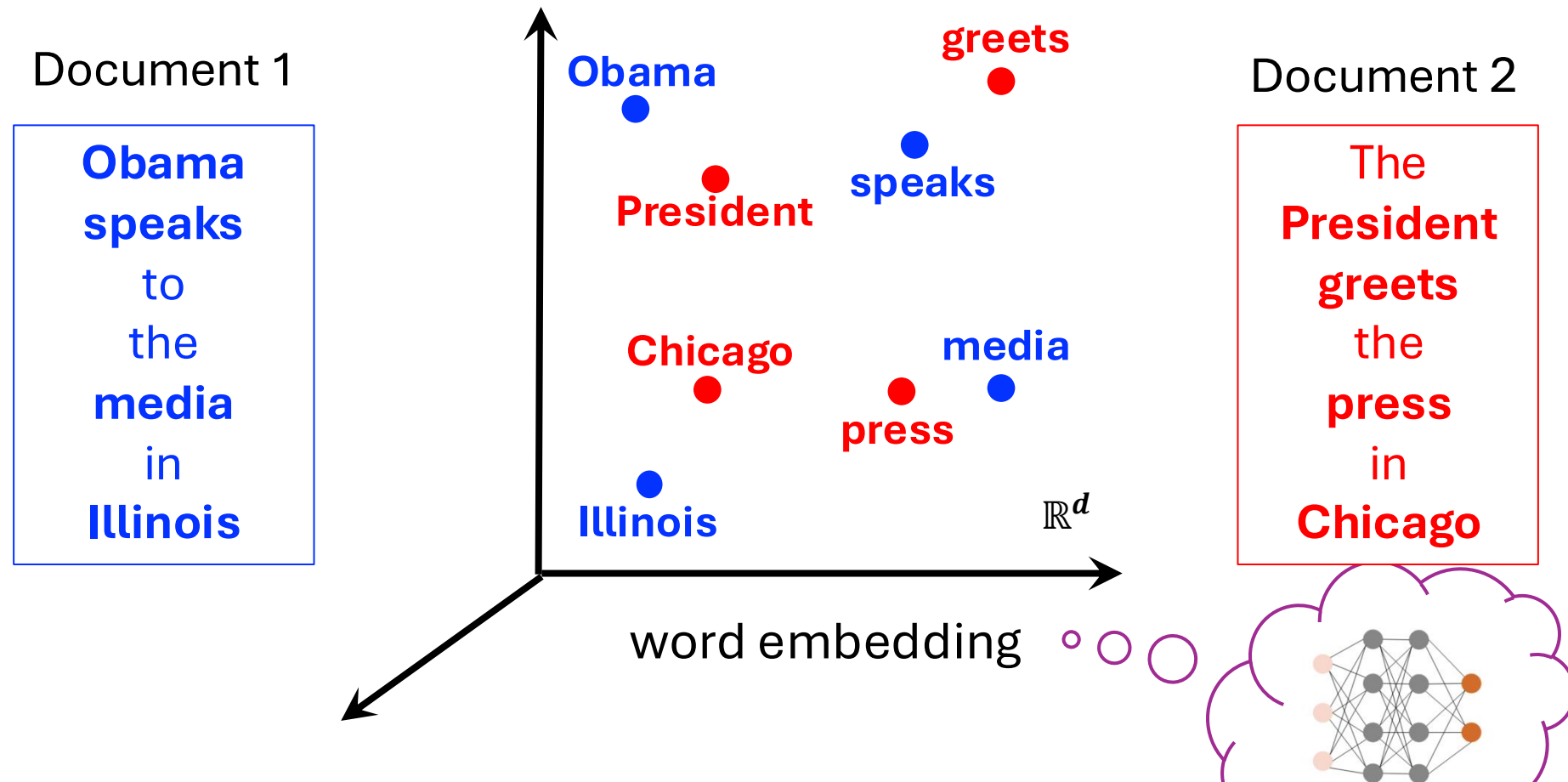
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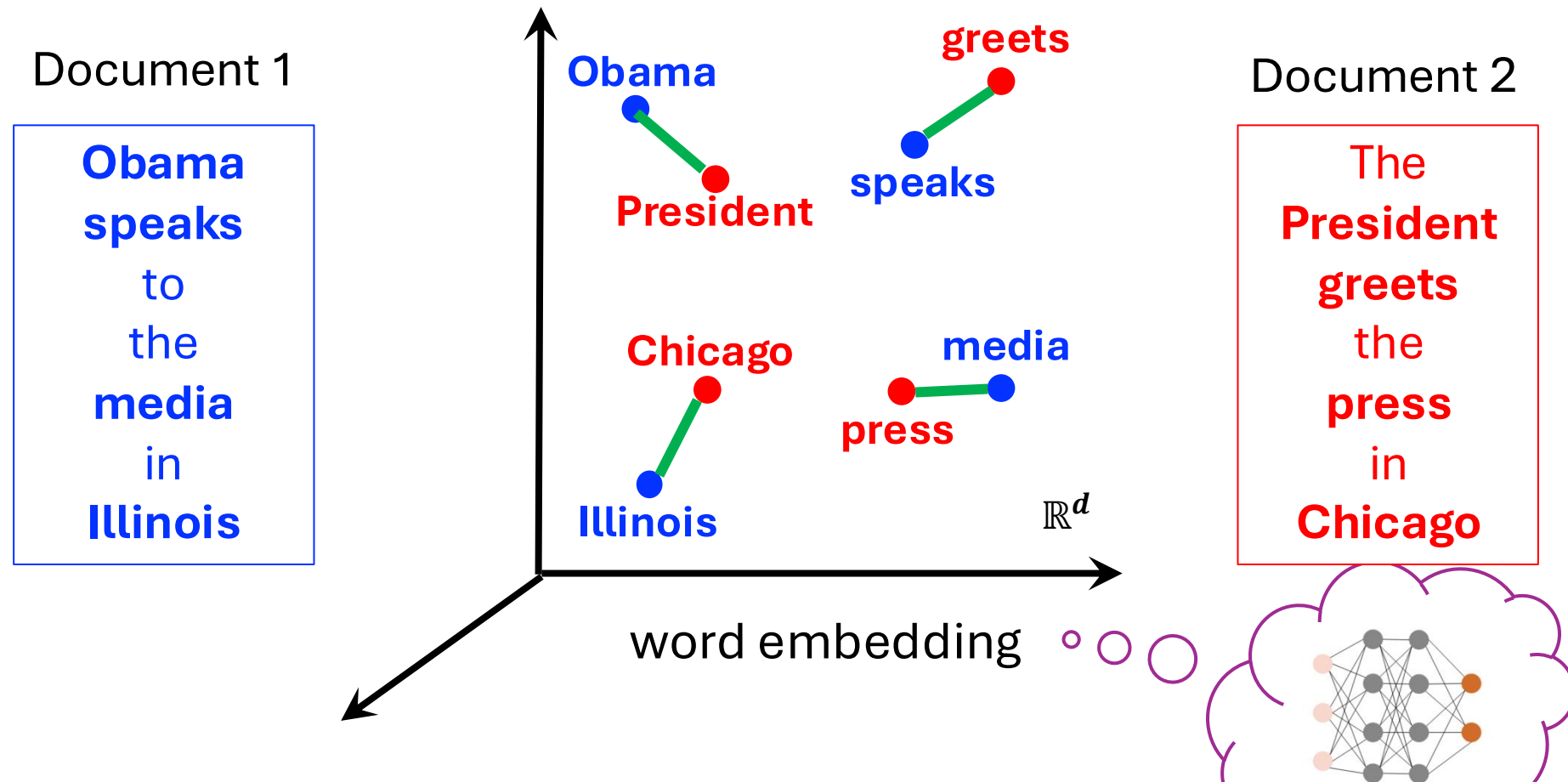
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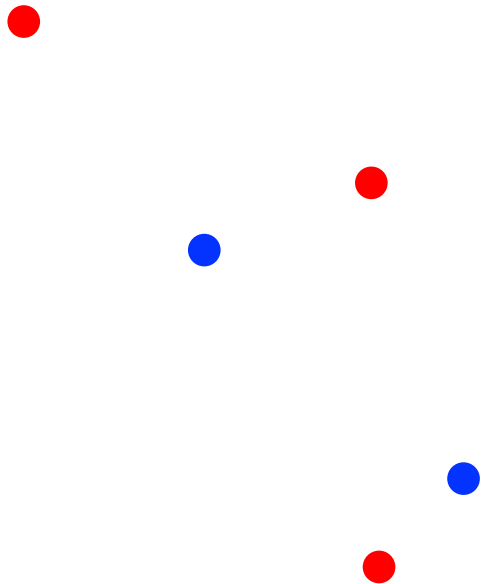
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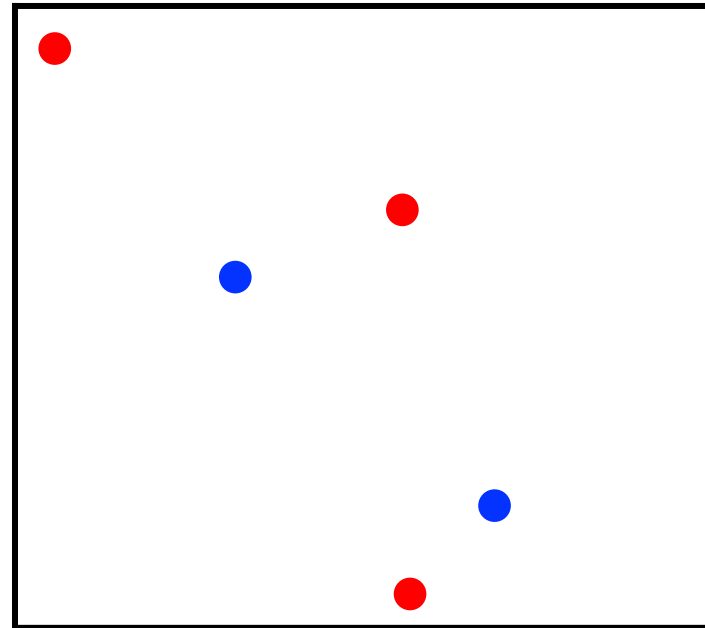
EMD Approximation

- **Theorem:** Suppose point coordinates are bounded in $[-\Delta, \Delta]$. Then, EMD can be approximated in near-linear time up to approximation $O(d \log(\Delta d))$ w.p. $2/3$.
- **Proof?**

EMD Approximation with Quadrees

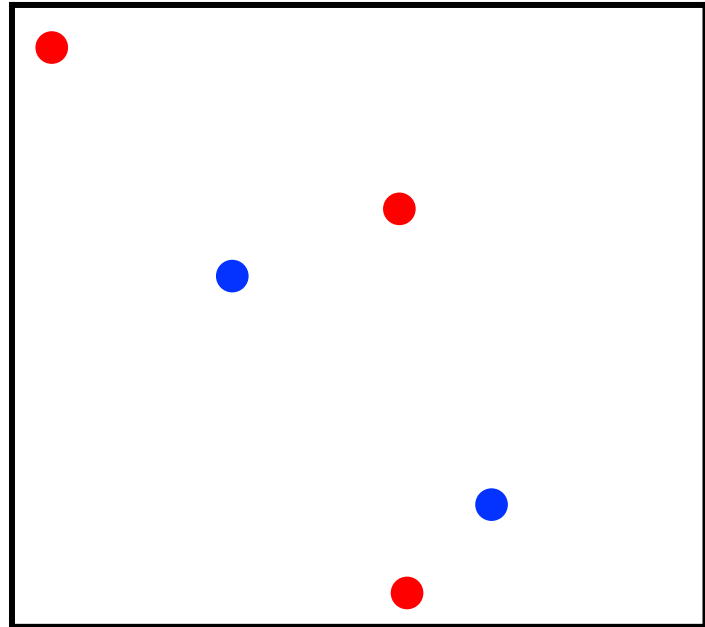


EMD Approximation with Quadrees

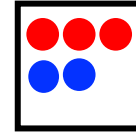


Side
length
 Δ

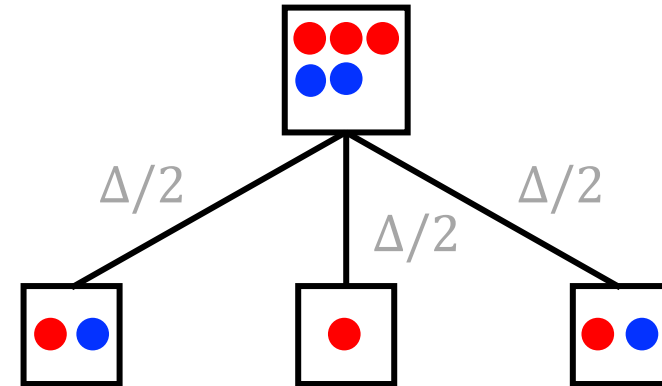
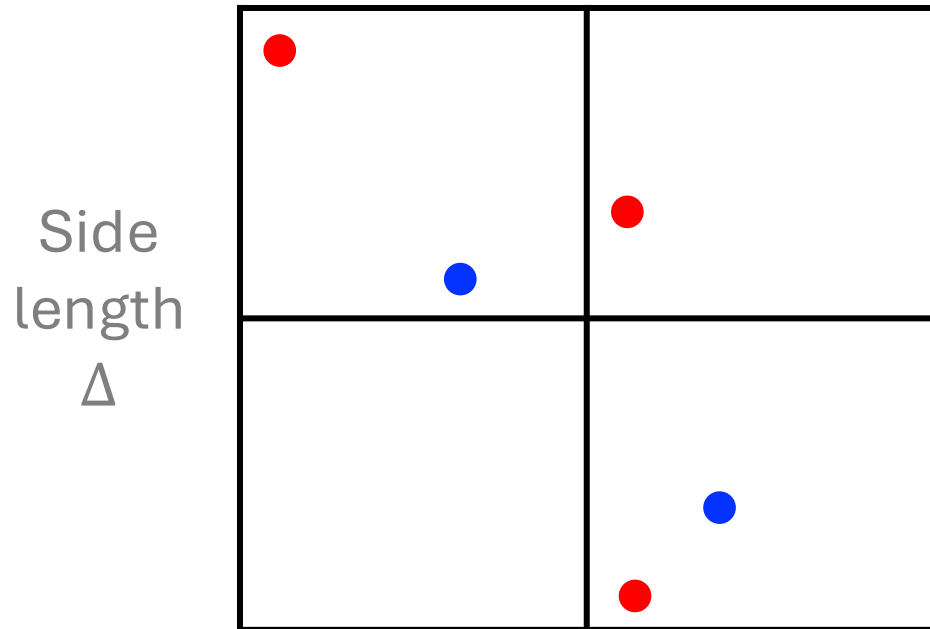
EMD Approximation with Quadtrees



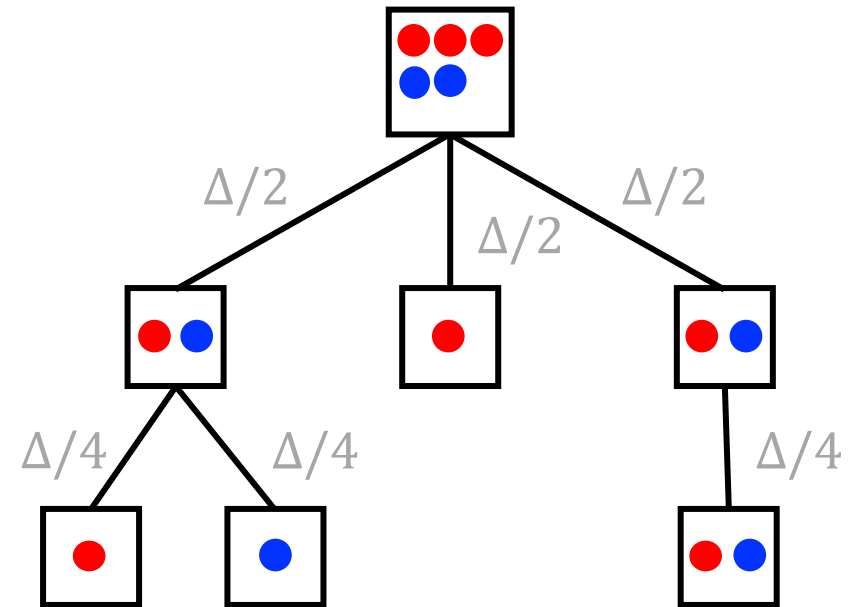
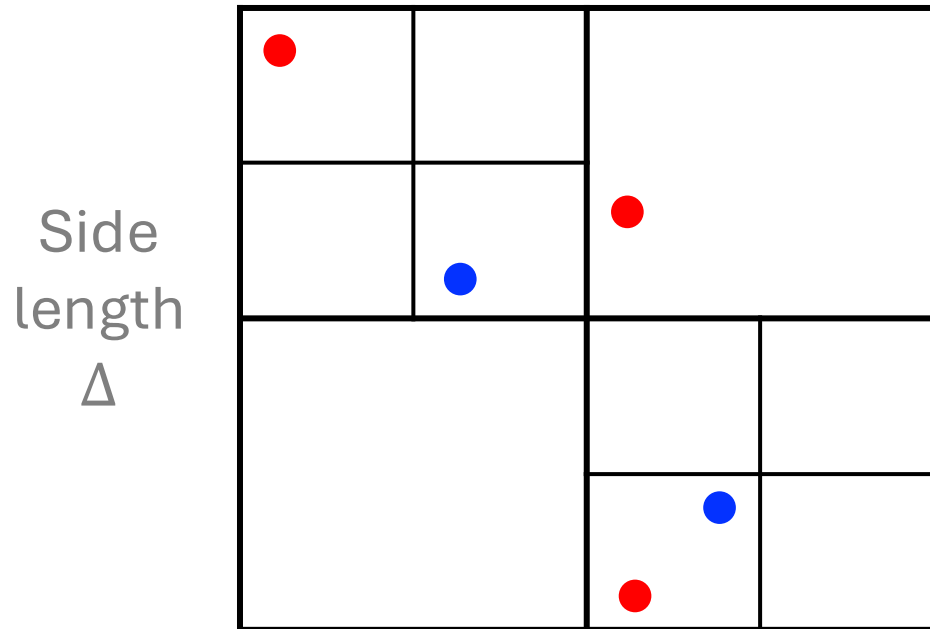
Side
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 Δ



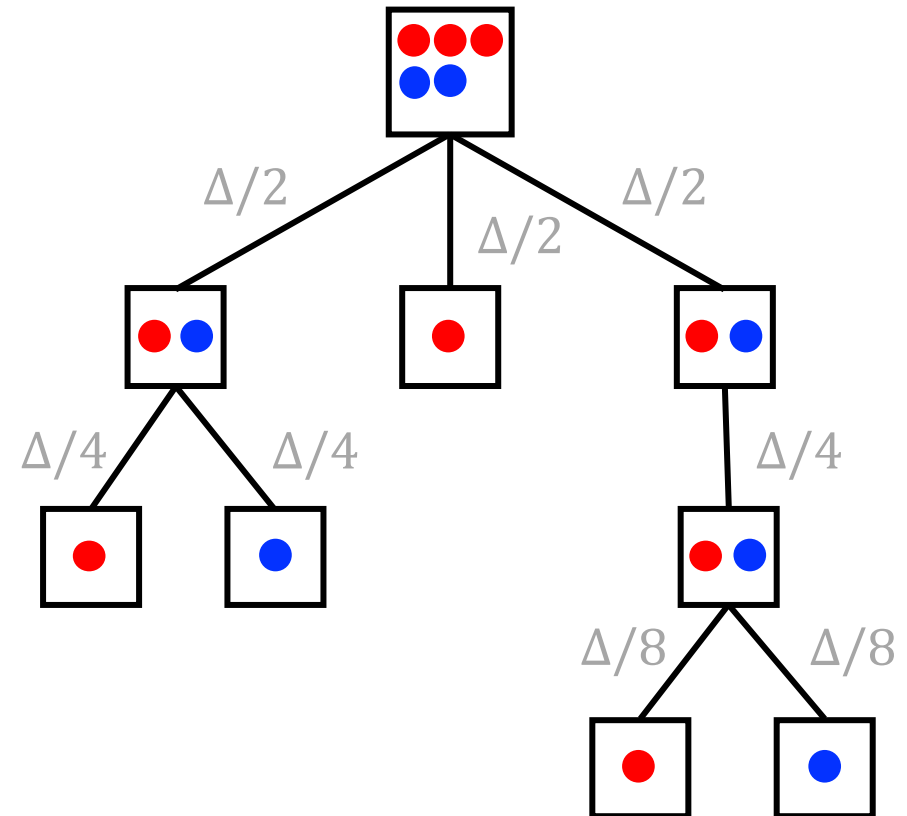
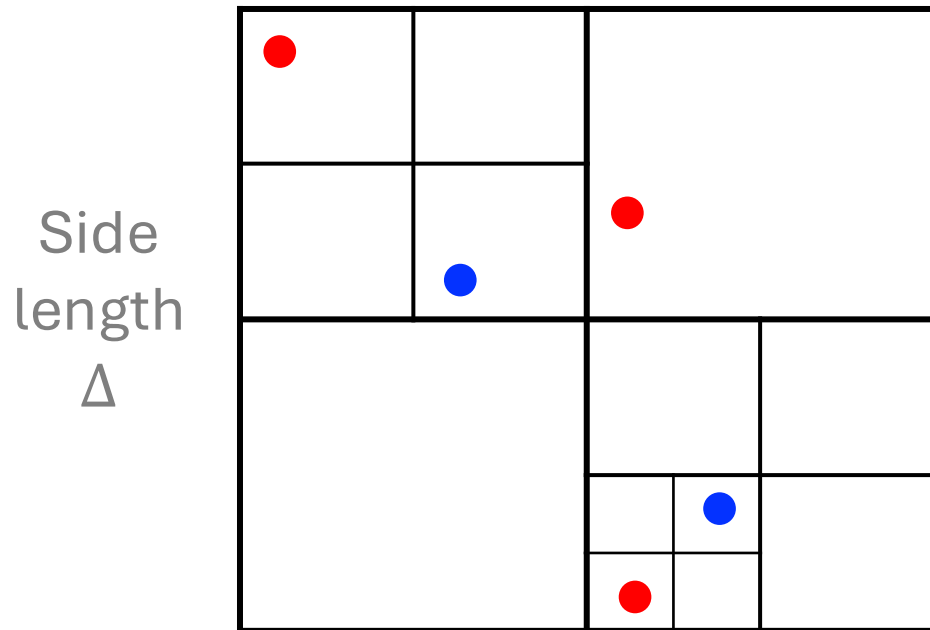
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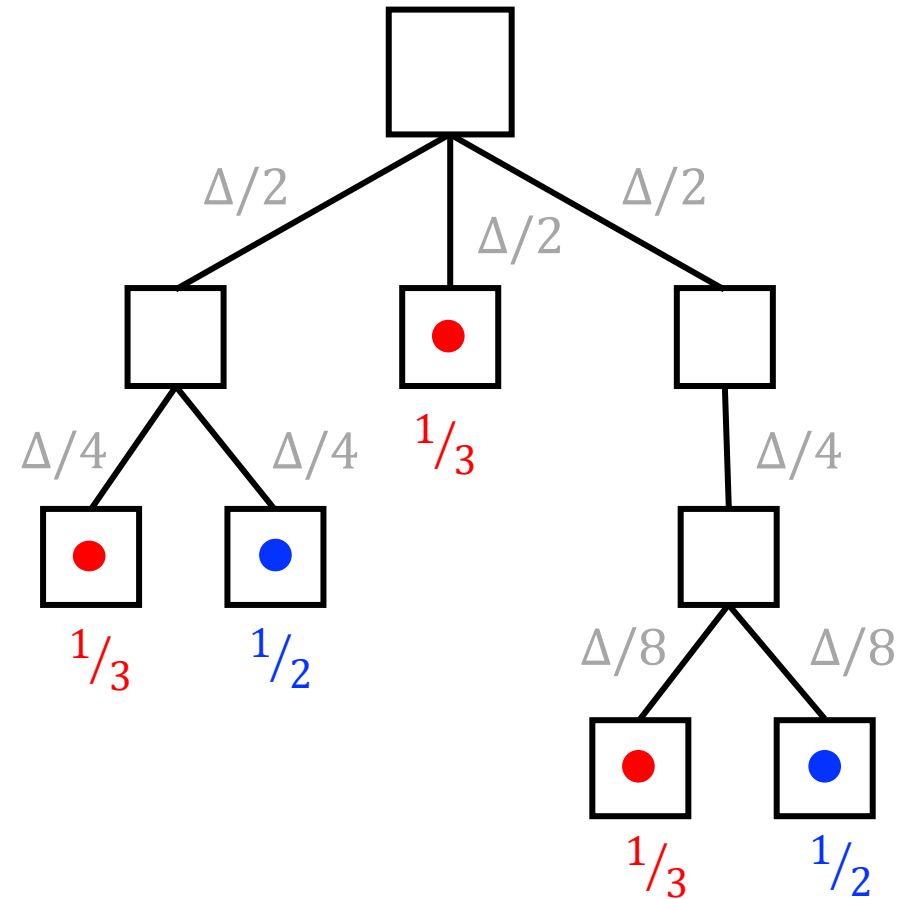
EMD Approximation with Quadrees



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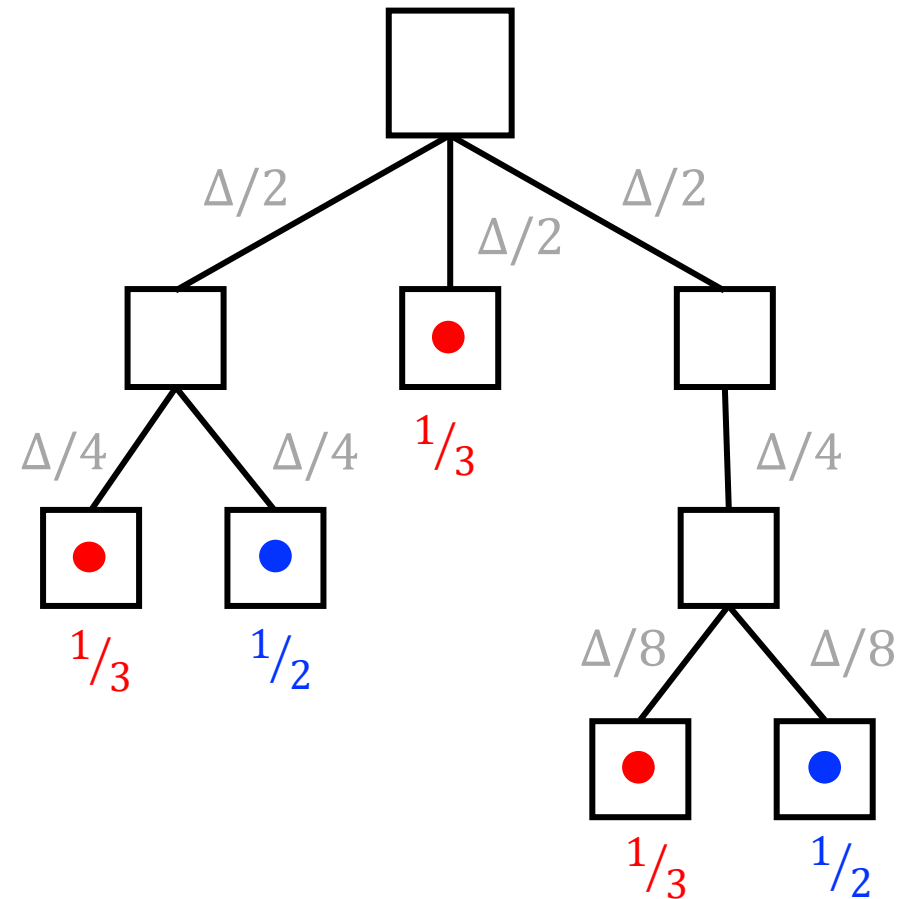


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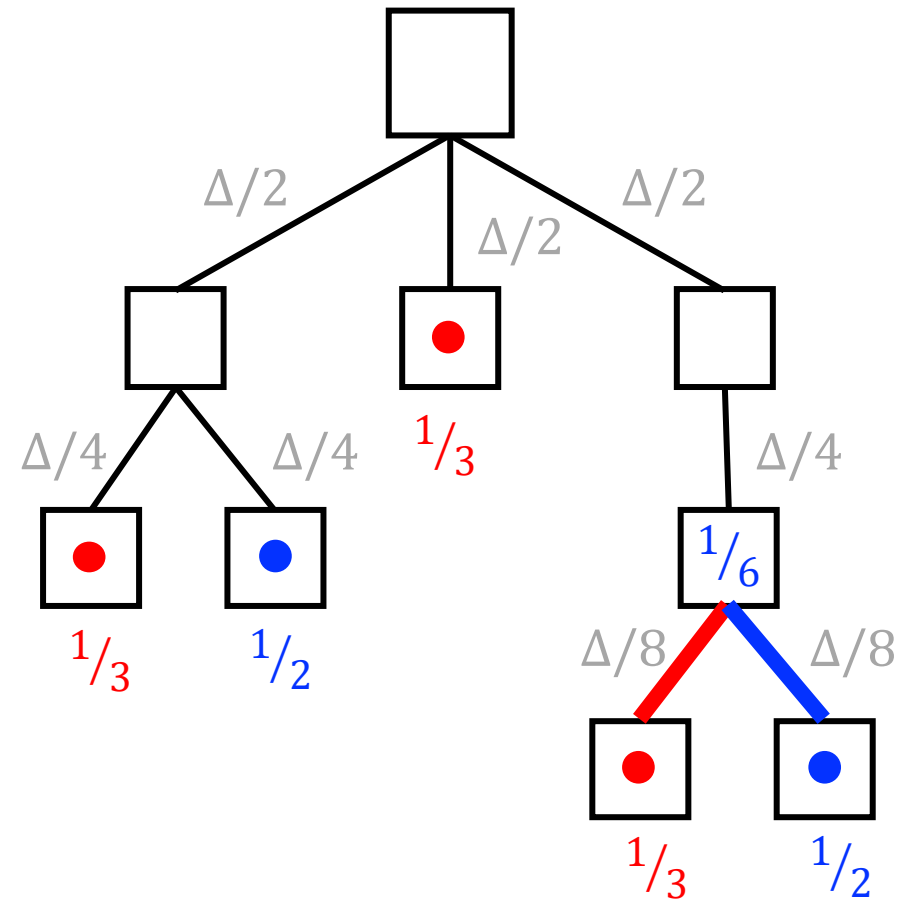
EMD Approximation with Quadrees

Compute: Optimal flow on tree



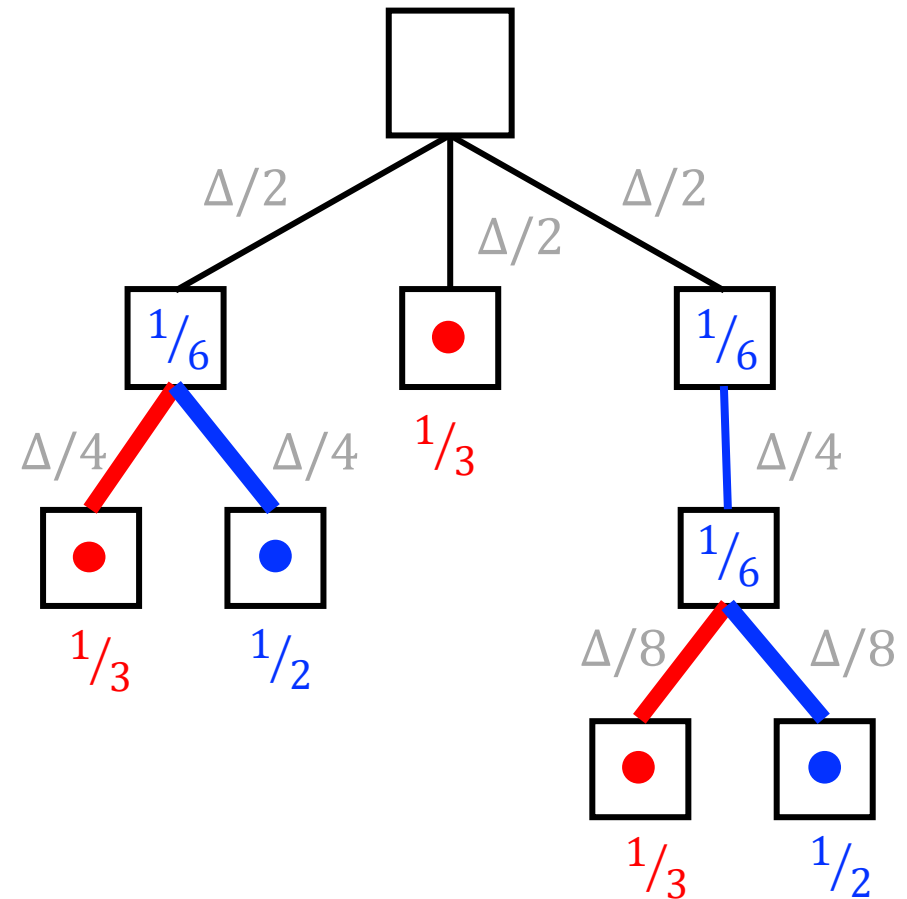
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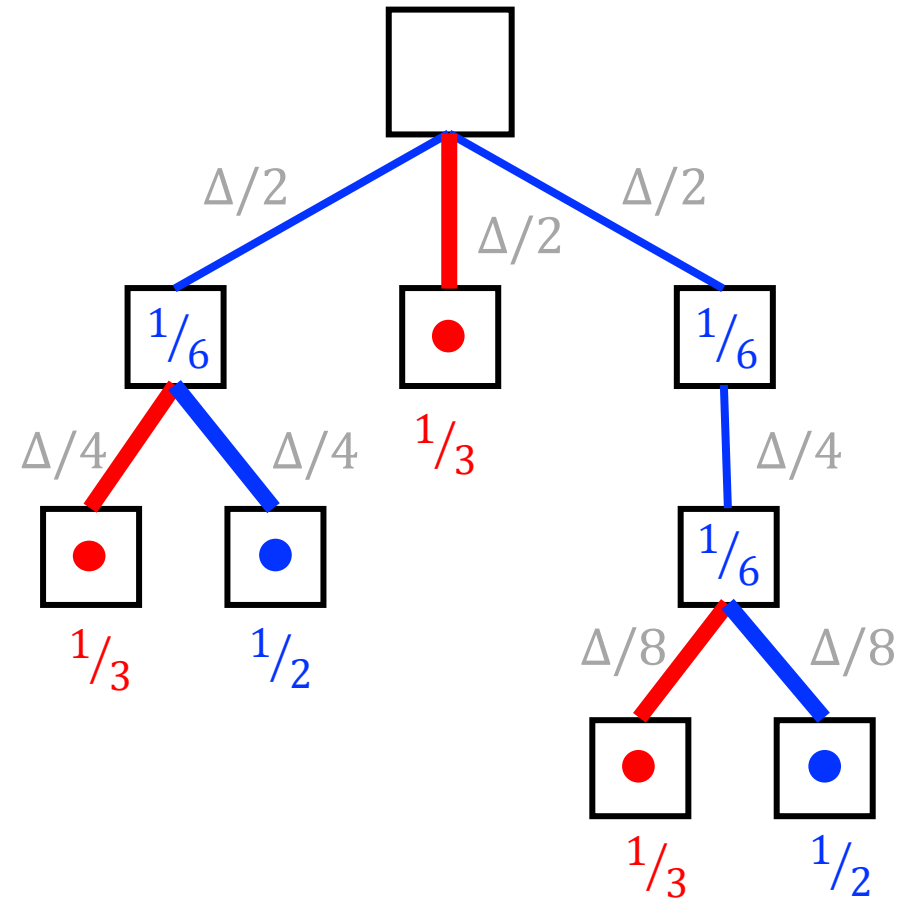
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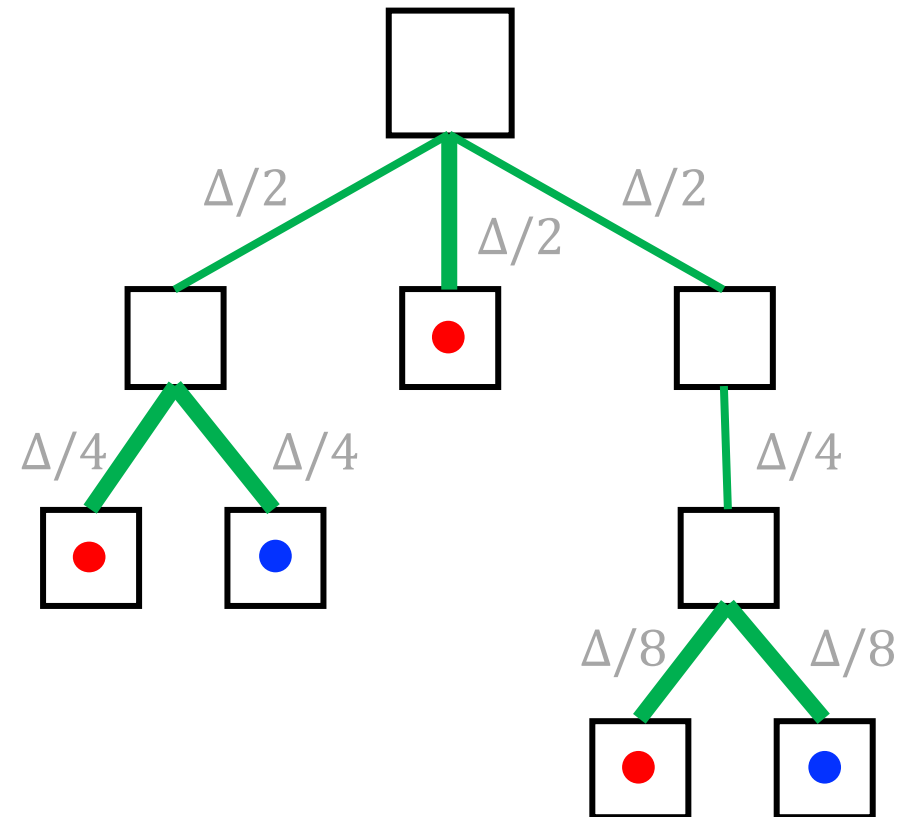


EMD Approximation with Quadrees

Compute: Optimal flow on tree

Return: EMD on tree metric

$$\sum_{\text{Tree edge } e} \text{weight}(e) \cdot \mathbf{F}_T(e)$$



EMD Approximation with Quadtrees

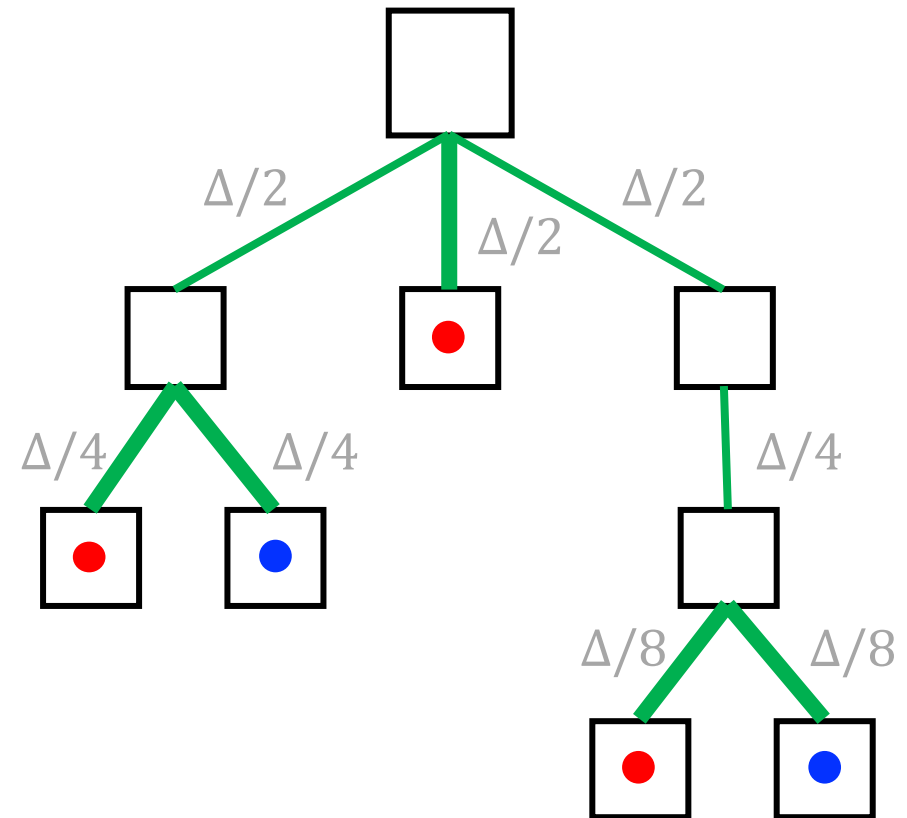
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$$= \sum_{x,y} T(x,y) \cdot \mathbf{F}_T(x,y)$$

$$= \text{EMD}_T(X, Y)$$



EMD Approximation with Quadrees

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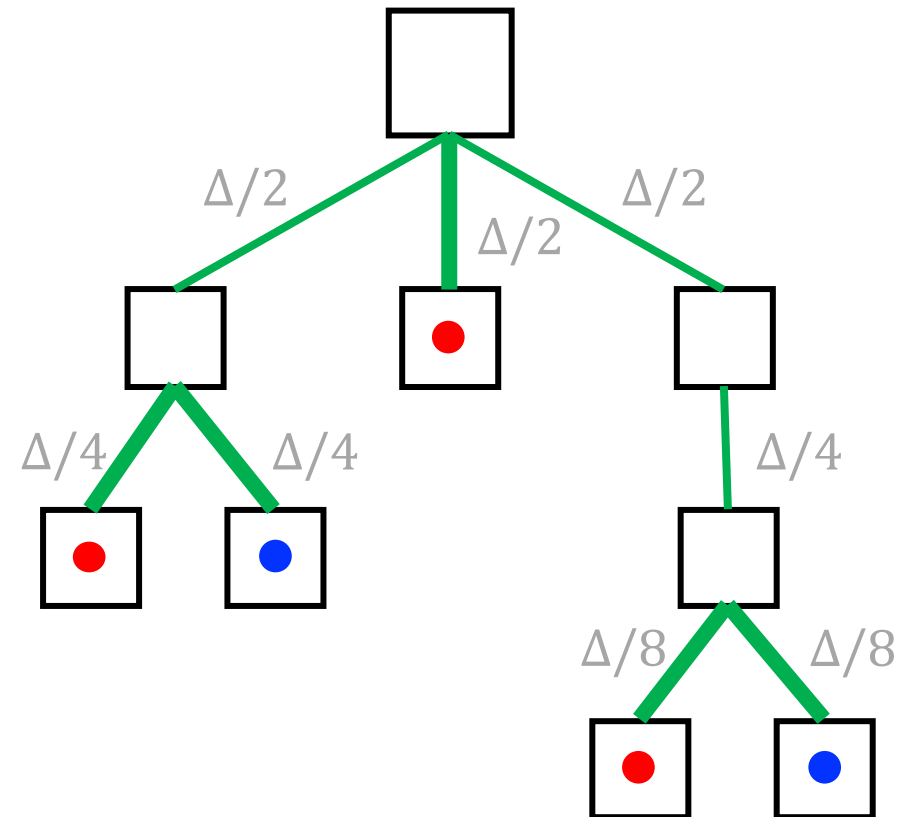
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EMD Approximation with Quadrees

- Recall our ℓ_1 distortion theorem: For every x, y ,
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Why?

Why?

Why?

Why?

Why?

What's the final step?

EMD Approximation with Quadrees

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Why?

Why?

Why?

Why?

Why?

And by Markov, we're done.



EMD Approximate Embedding into ℓ_1

- **Theorem:** The ℓ_1 -EMD metric over $\mathbb{R}^d$ with coordinates bound in $[-\Delta, \Delta]$ and min-distance 1 embeds into ℓ_1 with distortion $O(d \log(d\Delta))$.

EMD Approximate Embedding into ℓ_1

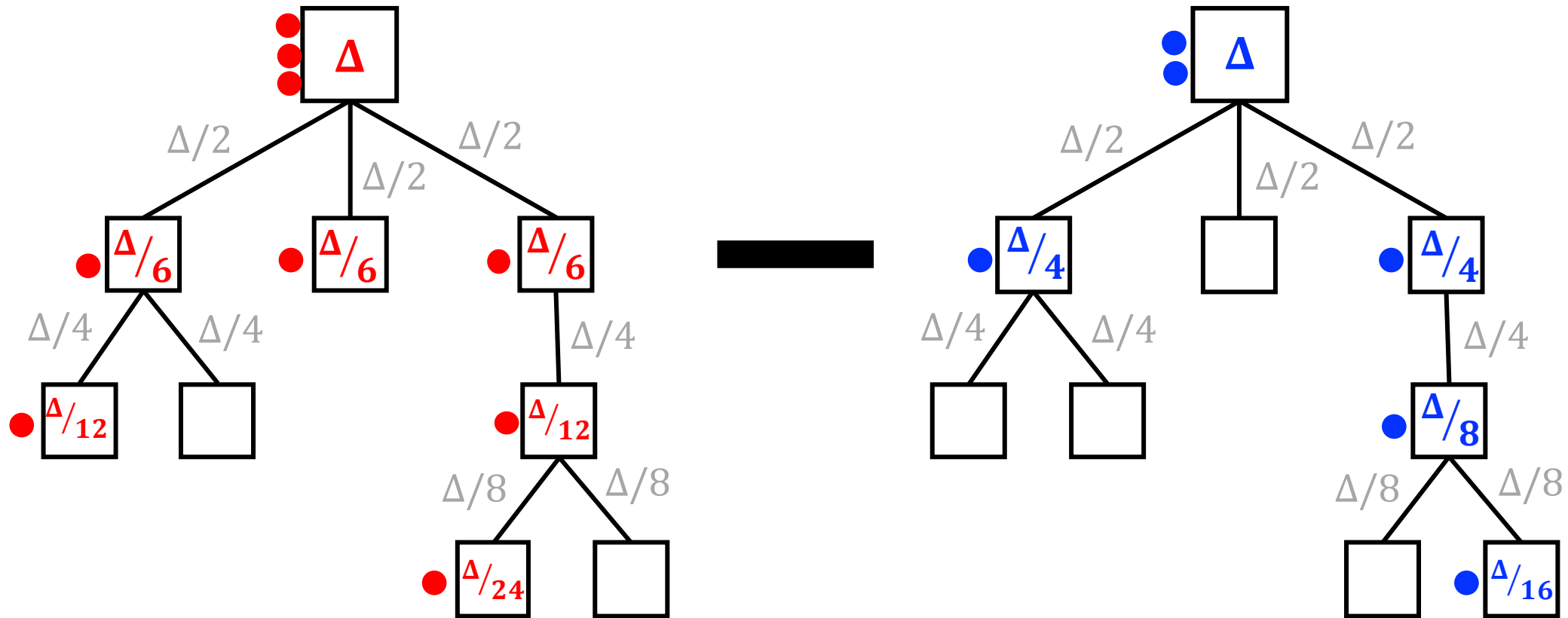
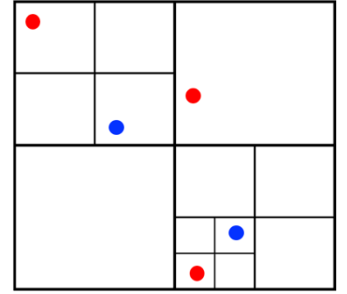
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 - **Observation:** Conditioned on a given quadtree, the tree-EMD distance is an ℓ_1 distance: $EMD_T(X, Y) = \|v_T(X) - v_T(Y)\|_1$ for some vectors $v(X), v(Y)$ **(how?)**

Tree-EMD as ℓ_1 Distance

Each tree level- ℓ tree node is a coordinate with value equal to mass of points in the corresponding hypercube times $2^{-\ell} \Delta$



EMD Approximate Embedding into ℓ_1

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 - We proved: $\Omega(d^{-1}) \cdot EMD(X, Y) \leq \mathbb{E}_T[EMD_T(X, Y)] \leq O(\log(\Delta d)) \cdot EMD(X, Y)$

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 - Hence: $\Omega(d^{-1}) \cdot EMD(X, Y) \leq \mathbb{E}_T[\|v_T(X) - v_T(Y)\|_1] \leq O(\log(\Delta d)) \cdot EMD(X, Y)$
 - But expectation is linear:

$$\mathbb{E}_T[\|v(X) - v(Y)\|_1] = \int_{\mathcal{T}} \Pr[T] \cdot \|v_T(X) - v_T(Y)\|_1 dT = \|V(X) - V(Y)\|_1$$

- Where $V(X)$ is the concatenation of the vectors $\Pr[T] \cdot v_T(X)$ for all possible trees T .



Nearest Neighbor Search in the EMD Metric

- Ideas how to achieve that given what we have seen?

Nearest Neighbor Search in the EMD Metric

- Ideas how to achieve that given what we have seen?
- [Indyk & Thaper '03]: Embed EMD into ℓ_1 with a quadtree, then use LSH for ℓ_1
 - ANN for EMD with sublinear query time!

EMD: Cardinality-Based Approximation

- We saw how to approximate ℓ_1 -EMD in near-linear time up to $O(d \log(\Delta d))$.

EMD: Cardinality-Based Approximation

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- Can we do better?

EMD: Cardinality-Based Approximation

- We saw how to approximate ℓ_1 -EMD in near-linear time up to $O(d \log(\Delta d))$.
- We started with a Euclidean metric and embedded into ℓ_1 , so we could assume by JL that $d \approx \log n$.
- But if our points are originally in ℓ_1 , there is no dimension reduction theorem, and d could be large.
- Can we do better?
- [Andoni, Indyk & Krauthgamer '08]: We can get $O(\log(\Delta d) \cdot \log n)$ approximation
 - Exponentially better in d , at the cost of only $\log n$
 - With the same quadtree algorithm!
 - The proof will be simple given everything we have proven so far

Modified Quadtree Analysis for EMD

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Modified Quadtree Analysis for EMD

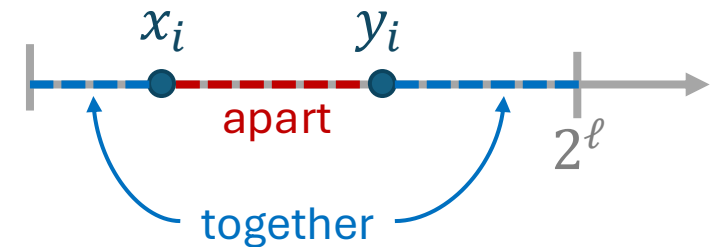
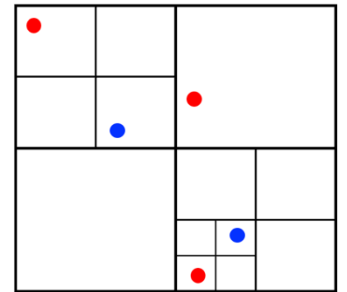
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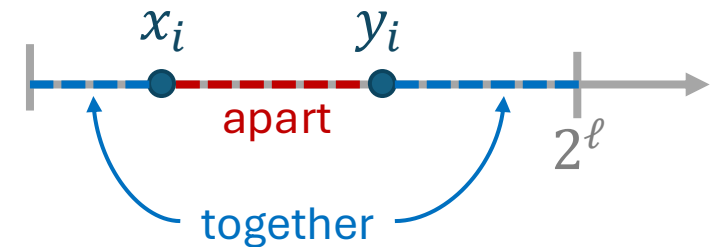
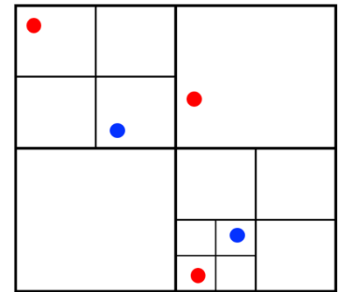
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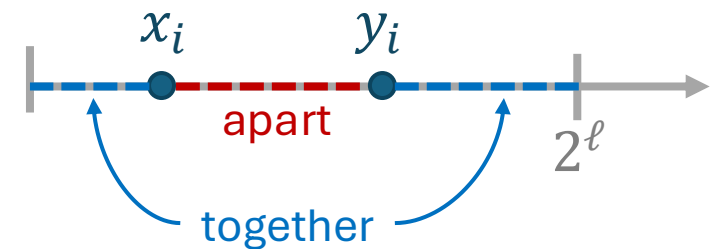
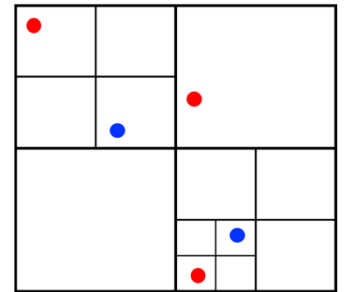
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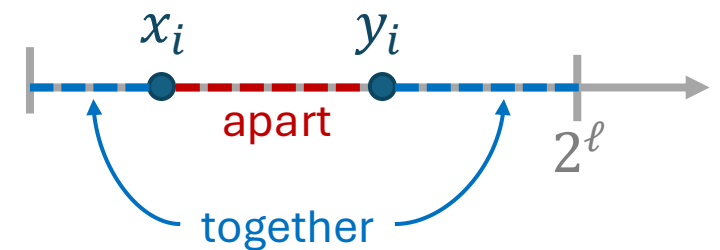
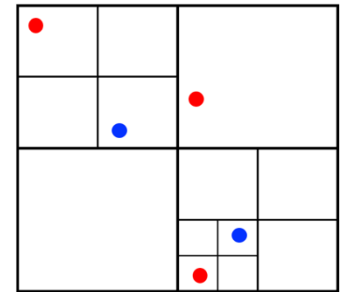
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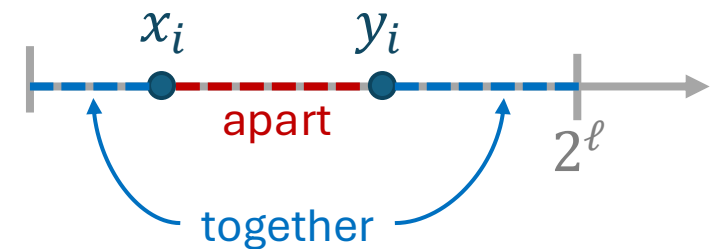
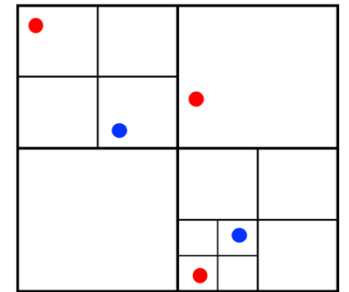
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 - By Markov, we have distortion $O(\log(\Delta d) \log n)$ for all pairs w.p. $2/3$.



Summary and Concluding Remarks

- We made the following assumptions: Points in $\mathbb{R}^d$, endowed with the ℓ_1 metric, with coordinates bounded in $[-\Delta, \Delta]$ and minimum distance 1
- Using quadtrees with a randomly shifted grid, we proved:
 - Probabilistic embedding of ℓ_1 metrics into tree metrics with distortion $O(d \log(\Delta d))$
 - ℓ_1 -ANN with near-linear build time and memory; polylog query and approximation
 - ℓ_1 -EMD embeds into ℓ_1 with distortion $O(d \log(\Delta d))$
 - Special case: EMD on the hypercube $\{0,1\}^d$ (with Hamming distance) embeds into ℓ_1 with distortion $O(d \log(d))$
 - [Khot & Naor '06]: The distortion is $\Omega(d)$
 - EMD approximation with distortion $O(\log(\Delta d) \cdot \min\{d, \log n\})$ in near-linear time
 - There are recent improvements to $\tilde{O}(\log n)$ -- will be in project-eligible papers

Questions?